Weighted Double-Low-Rank Decomposition With Application to Fabric Defect Detection

Dongmei Mo, Wai Keung Wong, Zhihui Lai, *Member, IEEE*, and Jie Zhou

***Abstract*— Recently, many methods based on low-rank representation have been proposed for fabric defect detection. Most of them relax the low-rank decomposition problem to a nuclear norm minimization (NNM) problem to pursue the convexity of the objective function. When solving the standard NNM problem, matrix singular values have to be treated equally. This, however, would be impractical in the scenario of fabric defect detection as the matrix singular values have clear physical meanings, and thus, they should be treated differently. In this article, we propose a weighted double-low-rank decomposition method (WDLRD) to treat the matrix singular values differently by assigning different weights. Thus, the most important/distinguishing characteristics of a fabric image can be preserved. Another difference between WDLRD and the other existing low-rank-based methods is that WDLRD considers a defective fabric image being decomposed to two low-rank matrices, i.e., low-rank defect-free matrix and low-rank defect matrix, as the defect-free and defective regions are usually composed of homogeneous objects that have a high correlation. Besides, WDLRD is more robust for defect detection in various situations by adding a noise term to avoid noise or other interference on the fabric surface. In addition, a defect prior is incorporated into the objective function of WDLRD to guide locating the defective regions. The proposed optimization problem can be easily solved by an iterative algorithm based on augmented Lagrange multipliers. Experimental results on TILDA, periodically patterned fabric, and Textile & Apparel Artificial Intelligence databases show that the proposed WDLRD obtains better performance than state-of-the-art methods in locating the defective regions on fabric images.**

***Note to Practitioners*—This article is motivated by the problem that the performance of fabric defect detection in the textile industry is poor. It is necessary to develop an effective method to improve the defect detection accuracy and reduce overall manufacturing cost. Existing automatic defect detection approaches usually contain two stages: first, capture fabric images from the weaving machine and then use a defect detection algorithm in a host computer to conduct a real-time inspection and give an alarm if defects occur. This article focuses on locating defects for given defective images after the procedure of binary classification (which determines an image as defective or defect free). The article proposes an objective function to mathematically interpret the optimization problem between fabric images and predictive defects. The optimal solution can be obtained by employing the alternating direction method of multipliers (ADMMs). The proposed method is described as a new defect detection algorithm. Extensive experiments were conducted to evaluate the algorithm, and the experimental results indicate that the proposed method is superior to many existing fabric defect detection methods. Preliminary experiments suggest that this method is feasible but has not yet been really used in production. In future research, we will collect more fabric images from the textile industry and develop large-scale databases for verifying the proposed method for real-life applications.**

***Index Terms*— Fabric defect detection, low-rank representation, patterned fabric, plain fabric, sparse matrix.**

I. Introduction

INSPECTION of fabric quality is an important process of fabric production in textile industry [1], [2]. As indicated in [3], more than 70 categories of fabric defects are established. Currently, the textile industry is heavily relying on human visual inspection methods, which results in unreliable inspection performance and thus increases overall manufacturing costs. Fabric with the presence of defects will lose 45%–65% of its original prices [4]. Therefore, it is necessary and reasonable to develop an automatic visual inspection technique to improve the overall performance and reliability of fabric inspection and, at the same time, improve the productivity and efficiency in the textile industry [5].

The existing automatic fabric defect detection methods can be roughly divided into three categories: spectral, statistical, and model-based methods [4]. Statistical methods find the distinct statistical texture characteristics by comparing different patches of the given fabric images. Existing statistical methods include histogram character analysis [6], local contrast enhancement [7], and the fractal method [8]. The main disadvantage of the statistical methods is that the performance of defect detection heavily depends on the size of the selected sliding window and the discrimination threshold. In addition, since they locate defective regions by comparing the difference between regions over the whole fabric image, they ignore the global information [9]. Spectral methods detect the surface defect regions by transforming the input images to the spectral

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domain and calculate the energy of the filter responses [10]. The spectral methods include Fourier [11], Gabor [12], and wavelet [13] transforms. Compared with statistical methods, spectral methods can make good use of global information on fabric images. Model-based methods solve the problem of defect detection via modeling and decomposing techniques. Both spectral and statistical methods can hardly inspect stochastic surface variation for a random textured fabric; however, model-based methods can obtain better detection accuracy in this case [14]. Among the model-based methods, low-rank decomposition is a research hotspot, and it has been applied to the real application of computer vision [15], [16]. In the scenario of defect detection, the background always has a visually homogeneous texture while any defects stand out from the background. Thus, the low-rank decomposition model can better fit for defect detection. For a given fabric image, it can be regarded as a combination of repeated patterns, especially for plain, twill, and regularly patterned fabrics. These repeated patterns are very correlated with each other and can be represented in low-rank fashion. However, defects destroy the regularity of these patterns and present as different and outstanding objects from the background [9]. Based on the characteristics of fabric defects, many low-rank decomposition methods are proposed to conduct defect detection [9], [10], [17], [18]. Chetverikov [19] used regular attributes of fabric to recover defect-free fabric and detect defects by residual of resonant singular value decomposition simultaneously. However, it obtained poor detection performance as fabric images usually suffer from fickle shadows and illumination variations, which would change or even destroy regular fabric attributes and finally result in a high false alarm. Robust principal component analysis (RPCA) [20] is widely used in many real applications of image completion, such as shadows and noise removing from face images and corrupted regions removing from building images. In addition, RPCA can be applied to fabric defect detection as it can recover background as a low-rank matrix and the defects as the sparse representation from a defective fabric image. Cao *et al.* [21] proposed modified RPCA with noise term and defect prior (PNRPCA) for fabric defect detection. PNRPCA is an RPCA-based method by adding a noise term to capture the Gaussian noise from fabric images. Cao *et al.* [22] also proposed another method using prior knowledge guided least squares regression (PGLSR) for fabric defect detection. The objective function of PGLSR is designed as the least square form by using L_2-norm to replace nuclear norm on the background matrix, and the optimization problem can be easily solved. However, such a strategy seems to be not suitable for the case when defects are presented in a long, tiny, or zigzag manner. In [10], a specific texture prior is designed based on the characteristics of fabric defects for low-rank-based automatic visual defect detection. The method can, to some extent, achieve low false alarm and high precision for fabric defect detection. Wang *et al.* [18] designed a randomized low-rank and sparse matrix decomposition model to recover the defects as a small portion of the whole image.

There are many other methods based on low-rank representation, and the optimization problems are usually solved by singular value soft-thresholding (SVT) methods [23]. However, using SVT to solve the standard nuclear norm optimization problem will shrink each singular value equally by subtracting the same constant, which means that all singular values are treated as equally important [24]. This, however, will miss prior knowledge that we often have from matrix singular values [24]. For example, for a given defective image, the defect-free fabric attributes are usually composed of the same textures or repeated patterns, and thus, the background matrix is supposed to be low rank. When recovering the low-rank background matrix, the larger singular values corresponding to major projection directions are supposed to be less shrunk to preserve the major components. Thus, we should preserve the most important information for the background matrix based on the observation of singular values. Furthermore, for the defect matrix, we can suppose that larger singular values corresponding to outstanding defect features should be less shrunk to locate potential defect regions. Unfortunately, existing methods do not consider such property and simply shrink each singular value by subtracting a constant.

In this article, we propose a novel method called weighted double-low-rank decomposition (WDLRD) to consider the prior knowledge on singular values of both background matrix and defect matrix for effective defect detection. The proposed WDLRD can be applied to the automatic fabric inspection system, which is presented in Fig. 1. The system architecture mainly contains two components, i.e., vision hardware and image processing software. The vision hardware component illuminates the woven fabric and captures the fabric images and sends them to the host computer, while the image processing component analyzes the images and gives corresponding outputs. Specifically, the inspection process is presented in Fig. 2. The charge-coupled device-based camera integrated with the weaving machine is used to capture the fabric images, and the images are sent to the control system installed on the host computer. Then, the control system uses the binary classifier to recognize the images as defective or defect free. Once a defective image is determined, it will be immediately sent to the proposed WDLRD to locate the defect regions, and the system will finally show corresponding outputs. The binary classifier in the image processing component can be easily implemented with the recent deep-learning technique, which obtains state-of-the-art feature extraction and classification performance. In this article, we do not analyze details of the binary classifier but focus on solving the problem that the defect detection performance in the textile manufacturing industry is poor. WDLRD performs pixel-level defect detection based on the assumption that the background of defective images are composed of highly correlated features, and thus, can be regarded as lying in a low-dimensional subspace. Meanwhile, the defects standing out from the background are also composed of correlated features that can be discovered as low rank. Besides, the objects that are different from background and defects can be considered as noises or perturbation [9]. Furthermore, a defect prior is constructed to guide WDLRD to locate the defect region by utilizing the difference between the defects and background of the fabric images. Experimental results show that when the prior is accurate, the proposed method can easily locate the defective regions while the

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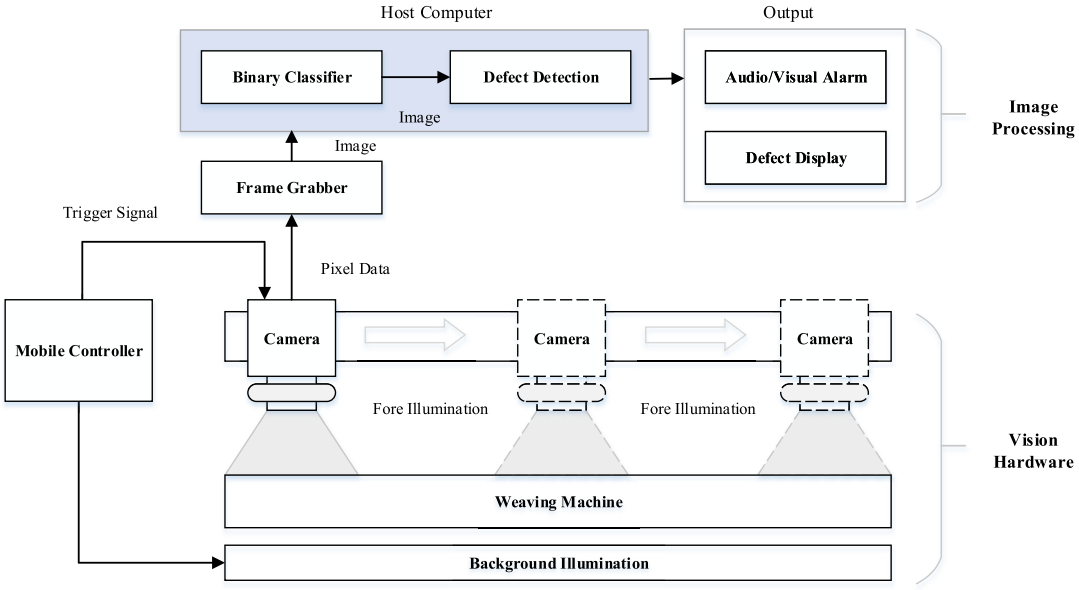


Fig. 1. Architecture of the automatic fabric inspection system. It contains two main components: vision hardware and image processing software. The vision hardware component is responsible for illuminating the woven fabric and capturing the images with the camera and sending them to the host computer, while the image processing component analyzes the images and gives corresponding outputs.

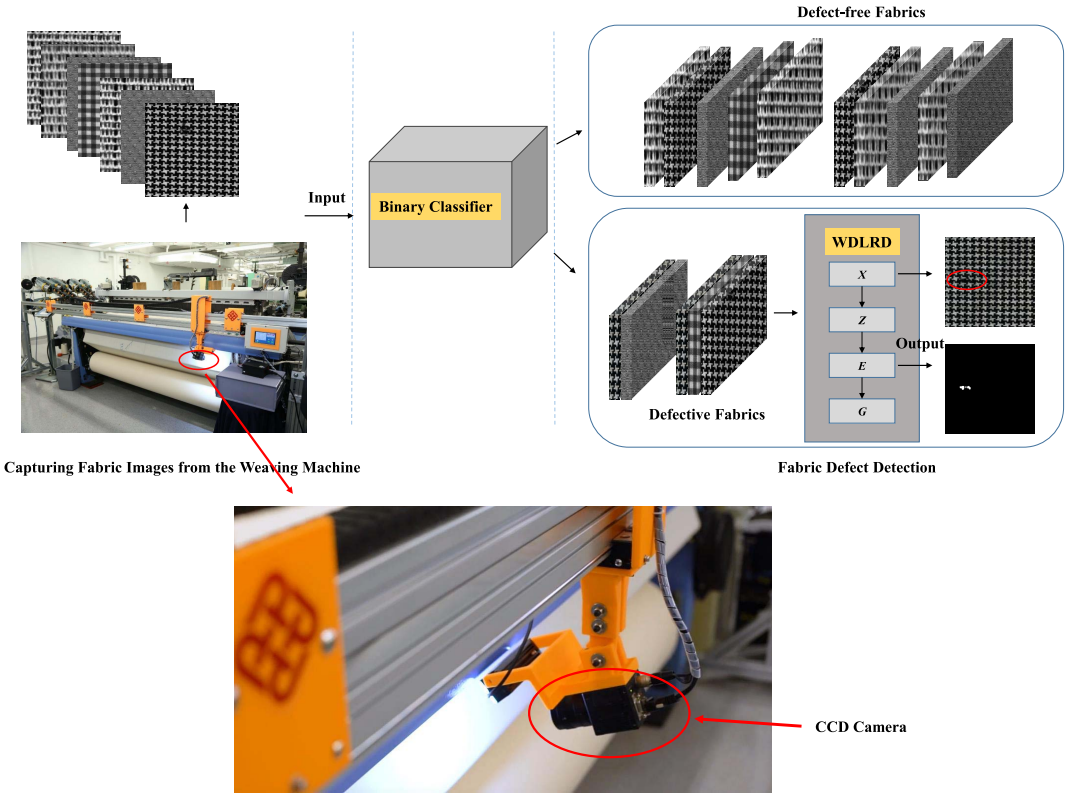


Fig. 2. Process of the image processing in the automatic fabric inspection system. The process contains three stages: first, capture the fabric image with the camera that integrated with the weaving machine; second, use the binary classifier to identity the input images as two categories: defective and defect free. Third, the proposed method (highlighted with a blue box in the lower right corner) is used to locate the defect region for the images that are recognized as defective.

defect prior is not very accurate, the proposed method can also automatically highlight the real defects and identify the fake defects. The contributions of this article can be mainly summarized as follows.

We propose a WDLRD method for fabric defect detection, which is the first time that a weighted double-low-rank technique is applied to fabric inspection. Different from existing low-rank-based defect detection methods, the proposed

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method can preserve the most important/distinguishing characteristics of a fabric image by treating the matrix singular values differently via assigning different weights. In addition, the proposed method locates defect regions based on the assumption that the defect-free and defective regions of a fabric image are usually composed of homogeneous objects that have a high correlation, and thus, they can both considered as low rank (i.e., double-low rank). By using such technique, we hope that the unnecessary information of the fabric image can be removed while the most important components that are corresponding to larger matrix singular values in both defect-free and defective regions can be preserved.

The remainder of this article is organized as follows. Section II briefly reviews the related methods, while Section III presents the proposed optimization problem and the corresponding optimal solution. Section IV shows experiments on three databases, and the conclusion is drawn in Section V.

II. Related Works

In this section, we first give the definition of the scalars and variables in this article and then briefly review several related works of the proposed method.

A. Notations

In this article, scalars are defined as lowercase or uppercase italic letters, such as i, j, k, and N. Vectors are denoted as bold lowercase italic letters, i.e., $\boldsymbol{x}$, $\boldsymbol{y}$, and so on, while matrices are represented as bold uppercase italic letters (i.e., $\boldsymbol{X}$ and $\boldsymbol{Y}$). $\|\boldsymbol{X}\|_2$, $\|\boldsymbol{X}\|_1$, and $\|\boldsymbol{X}\|_*$ denote the L_2-norm, L_1-norm, and nuclear norm of matrix $\boldsymbol{X}$, respectively. $\boldsymbol{A} \cdot \boldsymbol{B}$ represents the elementwise product of two matrices.

B. Robust PCA

Candès *et al.* [20] and Wright *et al.* [25] proposed RPCA to recover a low-rank matrix $\boldsymbol{Z}$ in order to maintain the basic information from corrupted images, and at the same time, obtain a sparse matrix $\boldsymbol{E}$ to capture noise or outliers. RPCA is a robust version of PCA that is more robust to outliers and gross corruption [20]. The solution of RPCA comes from solving the following problem:

$$\begin{aligned} &\min \|\boldsymbol{Z}\|_* + \lambda \|\boldsymbol{E}\|_1 \\ &\text{s.t. } \boldsymbol{X} = \boldsymbol{Z} + \boldsymbol{E}. \end{aligned} \tag{1}$$

where $\|\boldsymbol{Z}\|_* = \sum_i \sigma_i(\boldsymbol{Z})$ denotes the nuclear norm of matrix $\boldsymbol{Z}$, i.e., the sum of singular values of $\boldsymbol{Z}$, $\|\boldsymbol{E}\|_1 = \sum_{ij} |\boldsymbol{E}_{ij}|$ is L_1-norm penalty term, and λ is nonnegative parameter that balances the two terms. The optimal solution of RPCA can be obtained by efficient algorithms. RPCA, which obtains optimal solution for low-rank and sparse decomposition problem, can be applied to broad applications, such as face recognition, video surveillance, latent semantic indexing, ranking, collaborative filtering, and so on [26]. In the original article [25], iterative thresholding algorithm was proposed to solve the optimization problem of RPCA. However, it converges slowly [27]. To solve this problem, Lin *et al.* [27] proposed the augmented Lagrange multipliers (ALM) technique to obtain the optimal solution of RPCA. According to the exact ALM method in [27], the optimization problem in (1) can be rewritten as the following Lagrangian function:

$$\begin{aligned} L(\boldsymbol{Z}, \boldsymbol{E}, \hat{\boldsymbol{Y}}, \mu) = \|\boldsymbol{Z}\|_* + \lambda \|\boldsymbol{E}\|_1 + \langle \hat{\boldsymbol{Y}}, \boldsymbol{X} - \boldsymbol{Z} - \boldsymbol{E} \rangle \\ + \frac{\mu}{2} \|\boldsymbol{X} - \boldsymbol{Z} - \boldsymbol{E}\|_F^2 \end{aligned} \tag{2}$$

where $\hat{\boldsymbol{Y}}$ and $\mu > 0$ are multiplier and parameter for the deviation of the linear constraint of the convex optimization problem in (1), and $\langle \boldsymbol{A}, \boldsymbol{B} \rangle$ denotes standard trace of inner product of $\boldsymbol{A}$ and $\boldsymbol{B}$, respectively. The optimal $\hat{\boldsymbol{Z}}$ in (2) can be obtained by

$$\hat{\boldsymbol{Z}} = \arg\min_z \|\hat{\boldsymbol{Y}} - \boldsymbol{Z}\|_F^2 + \tau \|\boldsymbol{Z}\|_* \tag{3}$$

where τ is a positive constant, and we have

$$\hat{\boldsymbol{Z}} = \hat{\boldsymbol{U}} S_\tau(\hat{\boldsymbol{D}}) \hat{\boldsymbol{V}}^T \tag{4}$$

where $\hat{\boldsymbol{Y}} = \hat{\boldsymbol{U}} \hat{\boldsymbol{D}} \hat{\boldsymbol{V}}^T$ denotes the SVD of $\hat{\boldsymbol{Y}}$ and $S_\tau(\hat{\boldsymbol{D}})$ is soft-thresholding function defined as

$$S_\tau(\hat{\boldsymbol{D}})_{ii} = \max(\hat{\boldsymbol{D}}_{ii} - \tau, 0) \tag{5}$$

where $\hat{\boldsymbol{D}}_{ii}$ denotes diagonal element in $\hat{\boldsymbol{D}}$. In addition, many other nuclear norm minimization (NNM) methods (such as matrix completion [28], [29] and low-rank representation [30]) also used the above-mentioned SVT function to solve their optimization problems.

C. Modified RPCA

Cao *et al.* [21] proposed modified RPCA for fabric defect detection. The method assumes that when noise, fickle shadows, and illumination variations occur on the surface of fabric images, they would easily be regarded as defects, which will improve the false alarm in the application of fabric defect detection. Therefore, an additional noise term was added to (1) and the whole objective function of modified RPCA is as follows:

$$\begin{aligned} &\min \|\boldsymbol{Z}\|_* + \lambda \|\boldsymbol{E}\|_1 + \frac{\beta}{2} \|\boldsymbol{G}\|_F^2 \\ &\text{s.t. } \boldsymbol{X} = \boldsymbol{Z} + \boldsymbol{E} + \boldsymbol{G} \end{aligned} \tag{6}$$

where $\|\boldsymbol{G}\|_F^2$ is the square form of L_2-norm that characterize the Gaussian noise [31] and β is the balance parameter. The optimal solution of the above-mentioned optimization problem can also be obtained by the ALM technique.

D. WNNM-RPCA

Gu *et al.* [24] proposed weighted NNM (WNNM) to improve the flexibility of nuclear norm. The definition of WNNM is

$$\|\boldsymbol{X}\|_{w,*} = \sum_i |w_i \sigma_i(\boldsymbol{X})|_1 \tag{7}$$

where $\boldsymbol{w} = [w_1, w_2, \ldots, w_n](w_i) \geq 0$ is nonnegative weight assigned to $\sigma_i(\boldsymbol{X})$. Several different weight conditions were analyzed in [24] and theoretical analysis has presented that algorithm based on WNNM is as efficient as that based on NNM. More details about convergence of WNNM can be

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found in [24]. Gu *et al.* [32] integrated WNNM to RPCA and a new method called WNNM-RPCA was proposed. WNNM-RPCA aims at solving the following optimization problem:

$$\begin{aligned} &\min \|\boldsymbol{Z}\|_{w,*} + \lambda \|\boldsymbol{E}\|_1 \\ &\text{s.t. } \boldsymbol{X} = \boldsymbol{Z} + \boldsymbol{E}. \end{aligned} \tag{8}$$

By employing alternating direction method of multipliers (ADMMs), the problem in (8) can be rewritten as

$$L(\boldsymbol{Z}, \boldsymbol{E}, \bar{\boldsymbol{Y}}, \mu) = \|\boldsymbol{Z}\|_{w,*} + \lambda \|\boldsymbol{E}\|_1 + \langle \bar{\boldsymbol{Y}}, \boldsymbol{X} - \boldsymbol{Z} - \boldsymbol{E} \rangle + \frac{\mu}{2} \|\boldsymbol{X} - \boldsymbol{Z} - \boldsymbol{E}\|_F^2. \tag{9}$$

Similarly, the optimal $\bar{\boldsymbol{Z}}$ in (9) can be obtained by

$$\bar{\boldsymbol{Z}} = \arg\min_{z} \|\bar{\boldsymbol{Y}} - \boldsymbol{Z}\|_F^2 + \varphi \|\boldsymbol{Z}\|_{w,*} \tag{10}$$

where φ is a positive constant. Then, we have

$$\bar{\boldsymbol{Z}} = \bar{\boldsymbol{U}} S_{\frac{w}{2}}(\bar{\boldsymbol{D}}) \bar{\boldsymbol{V}}^T \tag{11}$$

where $\bar{\boldsymbol{Y}} = \bar{\boldsymbol{U}} \bar{\boldsymbol{D}} \bar{\boldsymbol{V}}^T$ denotes the SVD of $\bar{\boldsymbol{Y}}$ and $S_{(w/2)}(\bar{\boldsymbol{D}})$ is the soft-thresholding function with weight vector variable, that is,

$$S_{\frac{w}{2}}(\bar{\boldsymbol{D}})_{ii} = \max\left(\bar{\boldsymbol{D}}_{ii} - \frac{w_i}{2}, 0\right) \tag{12}$$

where $\bar{\boldsymbol{D}}_{ii}$ denotes diagonal element in $\bar{\boldsymbol{D}}$, w_i is the ith element in weight vector.

III. Proposed Method

In this section, we will first explain the motivations to improve defect detection performance and then present the optimization problem and the corresponding optimal solution for the proposed method.

A. Motivations

By analyzing and comparing RPCA, WNNM-RPCA with other defect detection methods based on low-rank representation, we have the following three motivations to design a new method to improve defect locating performance.

1) The common disadvantages of the RPCA and WNNM-RPCA are that they do not consider the prior knowledge of the defective images. As indicated in previous works [10], [21], [22], prior knowledge of the fabric image provides a helpful hint to discover the defects. As shown in Fig. 3(a) and (c), the defects present distinct textural characteristics from background of fabric images and prior map [Fig. 3(b) and (d)], thus, can highlight the potential detective regions.
2) Since RPCA and WNNM-RPCA do not focus on fabric image defect detection, they do not consider the specification and characteristics of fabric images. For example, from Fig. 3(a), we can see that fabric images always contain regular texture as background and irregular/outstanding textures as defects. Different pixels from the background have strong correlations, and the background matrix can be supposed to be low-rank. Similarly, for the defective regions, the pixels are also similar to each other, i.e., they also have a strong correlation with each other. Based on this observation, we can suppose that the defective regions are also low rank, and this can be more obvious when the defects are long and wide holes or blocks, as shown in Fig. 3(c). Therefore, it is reasonable to assume the background and defective regions are both low rank.
3) In real applications, fabric images are usually corrupted by not only defects but also noise, fickle shadows, and illumination variations. Under various conditions, even the same fabric can be presented somehow different. In such a case, the quality of automatic fabric defect detection cannot be guaranteed. Therefore, it is necessary to consider an additional noise term to capture the noise or other interference on the fabric images when designing a model for defect detection.

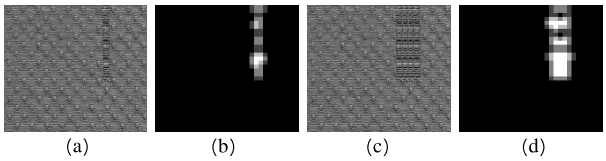


Fig. 3. Defective fabric images and defect priors. (a) and (c) are original defective fabric. (b) and (d) are defect priors.

B. Optimization Problem of WDLRD

Based on the analysis earlier, it is desirable to find out a solution that can effectively recover the background and discover the defects and capture the noise or interference from the defective images. Therefore, in this article, we propose a method called for fabric image defect detection. The proposed WDLRD aims to optimize the following problem:

$$\begin{aligned} &\|\boldsymbol{Z}\|_{w,*} + \alpha \|\boldsymbol{E}\|_{\tilde{w},*} + \beta \|\boldsymbol{G}\|_1 + \gamma \|\boldsymbol{W} \cdot \boldsymbol{E}\|_1 \\ &\text{s.t. } \boldsymbol{X} = \boldsymbol{Z} + \boldsymbol{E} + \boldsymbol{G} \end{aligned} \tag{13}$$

where $\boldsymbol{X} \in R^{d \times d}$ is the original image, $\boldsymbol{Z} \in R^{d \times d}$ is the background matrix, $\boldsymbol{E} \in R^{d \times d}$ and $\boldsymbol{G} \in R^{d \times d}$ are the defect matrix and noise matrix, respectively. Throughout this article, we assume the width and height of image are the same and d is the size of images. $\boldsymbol{W} \in R^{d \times d}$ is the weight matrix based on each pixel of the original image. α, β, and γ are the parameters to balance the four terms in (13).

In (13), the first two terms are penalized by WNNM. The assumption is that the pixels in the background or that in the defective regions, respectively, have a strong correlation, and thus, both the background matrix and the defect matrix should be low rank. In addition, since different matrix singular values have a different level of importance, they should be shrunk differently by subtracting different soft-thresholding values instead of a constant. The third term is L_1-norm-based penalty that aims to characterize the noise or illumination interference on fabric image surfaces. The fourth term is designed to guide defect detection by using defect prior. $\boldsymbol{W}$ is the weight matrix in which the position corresponding to smaller value indicates the possible defective position in the original image. The details of constructing $\boldsymbol{W}$ will be described in Section III-C.

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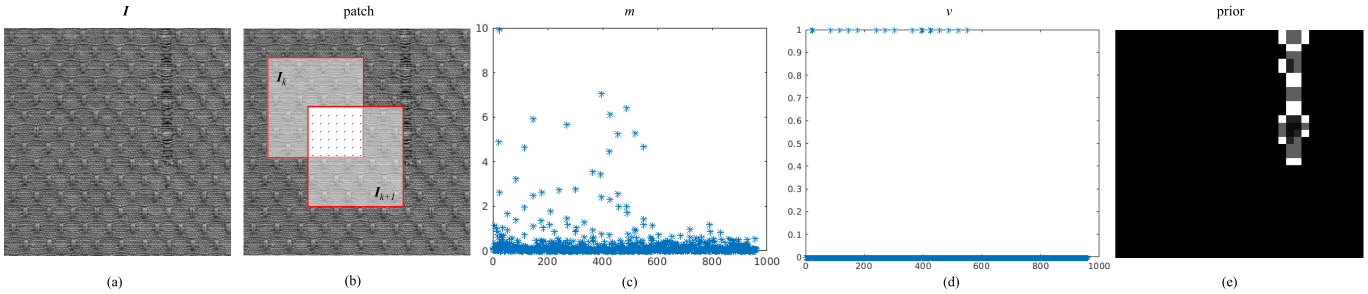


Fig. 4. Procedure of constructing prior map. (a) Input image $\boldsymbol{I}$. (b) Dividing image $\boldsymbol{I}$ to patch with patch size of $a \times a$ with $b \times b$ overlapping, and the overlapping area is highlighted with dot block. (c) Scatter of vector m. (d) Scatter of binary vector $\boldsymbol{v}$. (e) Final prior map: the darker region corresponding to larger values in $\boldsymbol{W}$ indicates the defect-free region, while the brighter region corresponding to smaller values in $\boldsymbol{W}$ indicates the defective region.

C. Construction of Defect Prior

As illustrated in [33], textons refer to fundamental microstructures in natural images (and videos) and are considered as the atoms of pre-attentive human visual perception. For an input fabric image, it can be viewed as a linear addition of some image bases selected from a base dictionary (such as Gabors and Laplacian of Gaussians). The base map can be further generated by a smaller number of textons selected from a texton dictionary [34]. For a given fabric image $\boldsymbol{I}$, we divide it into N patches by setting patch size as $a \times a$ with $b \times b$ overlapping. Then, for every pixel in each patch $\boldsymbol{I}_k$, we calculate the corresponding 8-D texton feature vector [34] as $\boldsymbol{t}_j$. The reason why we use 8-D texton feature is that the pixel-base-texton hierarchy presents an increasingly abstract visual description. The 8-D texton feature representation of each patch $\boldsymbol{I}_k$ is as follows:

$$\boldsymbol{x}_k = \frac{1}{\mid \boldsymbol{I}_k \mid} \sum_{p_j \in \boldsymbol{I}_k} \boldsymbol{t}_j \tag{14}$$

where $\boldsymbol{x}_k \in R^8$ is the average 8-D texton features for the kth patch, p_j is the jth pixel in patch $\boldsymbol{I}_k$, and $\mid \boldsymbol{I}_k \mid$ is the amount of pixels in patch $\boldsymbol{I}_k$. Then, we calculate the difference of each patch compared with the overall average patch features as

$$m_k = \left\| \boldsymbol{x}_k - \frac{1}{N} \sum_{i=1}^{N} \boldsymbol{x}_i \right\|_2 \tag{15}$$

where m_k is a scaler. Then, we have the a vector $\boldsymbol{v} \in R^N$ represents the difference of each patch, $\boldsymbol{v} = [m_1, m_2, \ldots, m_N]$. By using Otsu's method, we can learn a threshold T to convert the vector $\boldsymbol{v}$ to binary. That is,

$$\boldsymbol{v}_k = \begin{cases} 1, & \text{if } m_k > T \\ 0, & \text{otherwise.} \end{cases} \tag{16}$$

Then, we have the following matrix $\boldsymbol{M}$ representing the difference of each pixel in the fabric image $\boldsymbol{I}$

$$\boldsymbol{M}_{i,j} = \boldsymbol{v}_k, \quad \text{if } \boldsymbol{I}_{i,j} \in \boldsymbol{I}_k. \tag{17}$$

For the overlapping pixels, the corresponding values in $\boldsymbol{M}_{i,j}$ is the sum of the values in $\boldsymbol{v}$ that are overlapping. That is,

$$\boldsymbol{M}_{i,j} = \boldsymbol{v}_k + \boldsymbol{v}_h, \quad \text{if } \boldsymbol{I}_{i,j} \in \boldsymbol{I}_k \cap \boldsymbol{I}_h. \tag{18}$$

Finally, the defect prior $\boldsymbol{W}$ is

$$\boldsymbol{W}_{i,j} = \exp(-\boldsymbol{M}_{i,j}) \tag{19}$$

where smaller $\boldsymbol{W}_{i,j}$ indicates higher probability to be defective. The procedure of constructing the prior map is shown in Fig. 4. As we can see from Fig. 4, the input image $\boldsymbol{I}$ can be divided into N patches with size of $a \times a$ (the red bounding box) with $b \times b$ overlapping [the highlighted dot block in Fig. 4(b)]. The difference of each patch versus overall patch average is shown in Fig. 4(c), while the binary distribution of the patches is shown in Fig. 4(d). The final defect prior $\boldsymbol{W}$ is presented in Fig. 4(e) in which the brighter region corresponding to smaller values indicates higher possibility of defective region.

D. Optimization of WDLRD

The optimization problem of (13) is equal to

$$\begin{aligned} &\|\boldsymbol{Z}\|_{w,*} + \alpha \|\boldsymbol{E}\|_{\tilde{w},*} + \beta \|\boldsymbol{G}\|_1 + \gamma \|\boldsymbol{W} \cdot \boldsymbol{P}\|_1 \\ &\text{s.t. } \boldsymbol{X} = \boldsymbol{Z} + \boldsymbol{E} + \boldsymbol{G}, \quad \boldsymbol{E} = \boldsymbol{P}. \end{aligned} \tag{20}$$

The reason why we rewrite (13)–(20) by adding a new matrix variable $\boldsymbol{P}$ is that in such way the optimization problem can be solved iteratively by separating each term. Then, for (20), its augmented Lagrangian function is

$$\begin{aligned} &L(\boldsymbol{Z}, \boldsymbol{E}, \boldsymbol{G}, \boldsymbol{P}, \boldsymbol{Y}, \mu) \\ &\quad = \|\boldsymbol{Z}\|_{w,*} + \alpha \|\boldsymbol{E}\|_{\tilde{w},*} + \beta \|\boldsymbol{G}\|_1 \\ &\qquad + \gamma \|\boldsymbol{W} \cdot \boldsymbol{P}\|_1 + \frac{\mu}{2} \left\| \boldsymbol{X} - \boldsymbol{Z} - \boldsymbol{E} - \boldsymbol{G} + \frac{\boldsymbol{Y}_1}{\mu} \right\|_F^2 \\ &\qquad + \frac{\mu}{2} \left\| \boldsymbol{E} - \boldsymbol{P} + \frac{\boldsymbol{Y}2}{\mu} \right\|_F^2. \end{aligned} \tag{21}$$

Since the optimization problem in (21) is unconstrained, it can be minimized with respect to $\boldsymbol{Z}$, $\boldsymbol{E}$, $\boldsymbol{G}$, and $\boldsymbol{P}$, respectively, by fixing other variables [28].

1) Updating $\boldsymbol{Z}$: By fixing other variables, the optimal solution of $\boldsymbol{Z}$ can be obtained by minimizing the following problem:

$$\boldsymbol{Z}^* = \arg\min_{z} \|\boldsymbol{Z}\|_{w,*} + \frac{\mu}{2} \left\| \boldsymbol{Z} - (\boldsymbol{X} - \boldsymbol{E} - \boldsymbol{G} + \frac{\boldsymbol{Y}_1}{\mu}) \right\|_F^2. \tag{22}$$

Compared with the convex optimization problem in (1), the optimization problem of $\boldsymbol{Z}$ is not convex in general, and

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thus, it cannot be solved directly. Gu *et al.* [24] have analyzed the optimal solution to the objective function in the form of WNNM as in (22). NNM is proven to be a special case of WNNM when weights are set as a constant in WNNM [34]. As we can see, the form of the optimization problem (22) is similar to that of (10). Therefore, let $\boldsymbol{Y} = \boldsymbol{X} - \boldsymbol{E} - \boldsymbol{G} + (\boldsymbol{Y}_1/\mu)$ and $\boldsymbol{Y} = \boldsymbol{U}\boldsymbol{D}\boldsymbol{V}^T$ is the SVD of $\boldsymbol{Y}$, then we have

$$\boldsymbol{Z}^* = \boldsymbol{U} S_{\frac{w}{2}}(\boldsymbol{D}) \boldsymbol{V}^T \tag{23}$$

where

$$S_{\frac{w}{2}}(\boldsymbol{D})_{ii} = \max\left(\boldsymbol{D}_{ii} - \frac{w_i}{2}, 0\right). \tag{24}$$

The weights assigned to different singular values $\boldsymbol{D}_{ii}$ are designed as

$$w_i = \frac{C}{\sigma_i(\boldsymbol{Z}) + \eta} \tag{25}$$

where $\sigma_i(\boldsymbol{Z})$ denotes the ith singular value of $\boldsymbol{Z}$, $C > 0$ is a constant, and $\eta = 10^{-6}$. The idea in (25) is that in fabric defect detection, the larger singular values of the matrix indicate the more important components, and thus, they should be less shrunk. Therefore, the weight assigned to $\boldsymbol{D}_{ii}$ should be inversely proportional to $\sigma_i(\boldsymbol{Z})$ [24].

2) Updating $\boldsymbol{E}$: Similarly, the optimal solution of $\boldsymbol{E}$ can be obtained by fixing other variables and then we have

$$\boldsymbol{E}^* = \arg\min_E \alpha \|\boldsymbol{E}\|_{\tilde{w},*} + \frac{\mu}{2}\left\|\boldsymbol{E} - (\boldsymbol{X} - \boldsymbol{Z} - \boldsymbol{G} + \frac{\boldsymbol{Y}_1}{\mu})\right\|_F^2 + \frac{\mu}{2}\left\|\boldsymbol{E} - (\boldsymbol{P} - \frac{\boldsymbol{Y}_2}{\mu})\right\|_F^2. \tag{26}$$

Let $\boldsymbol{A} = \boldsymbol{X} - \boldsymbol{Z} - \boldsymbol{G} + (\boldsymbol{Y}_1/\mu)$, $\boldsymbol{B} = \boldsymbol{P} - (\boldsymbol{Y}_2/\mu)$, and the optimization problem in (26) can be rewritten as

$$\boldsymbol{E}^* = \arg\min_E \alpha \|\boldsymbol{E}\|_{\tilde{w},*} + \frac{\mu}{2}\|\boldsymbol{E} - \boldsymbol{A}\|_F^2 + \frac{\mu}{2}\|\boldsymbol{E} - \boldsymbol{B}\|_F^2. \tag{27}$$

That is,

$$\boldsymbol{E}^* = \arg\min_E \alpha \|\boldsymbol{E}\|_{\tilde{w},*} + \frac{\mu}{2}\left\|\boldsymbol{E} - \frac{1}{2}(\boldsymbol{A} + \boldsymbol{B})\right\|_F^2. \tag{28}$$

Let $\boldsymbol{H} = (1/2)(\boldsymbol{A} + \boldsymbol{B})$ and $\boldsymbol{H} = \tilde{U}\tilde{D}\tilde{V}^T$, then we have

$$\boldsymbol{E}^* = \tilde{U} S_{\frac{\tilde{w}}{2}}(\tilde{D}) \tilde{V}^T \tag{29}$$

where

$$S_{\frac{\tilde{w}}{2}}(\tilde{D})_{ii} = \max\left(\tilde{D}_{ii} - \frac{\tilde{w}_i}{2}, 0\right). \tag{30}$$

The weights assigned to different singular values $\tilde{D}_{ii}$ are designed as

$$\tilde{w}_i = \frac{C}{\sigma_i(\boldsymbol{E}) + \eta} \tag{31}$$

where $\sigma_i(\boldsymbol{E})$ denotes the ith singular value of $\boldsymbol{E}$. Similarly, the optimal solution to problem (31) can be obtained in a similar way as in [34].

Algorithm 1 Defect Detection by WDLRD

Input: Fabric image $\boldsymbol{X}$, defect prior $\boldsymbol{W}$, parameter α, β, γ.
Initialize $\mu = e^{-6}, \varepsilon = e^{-6}, maxStep = e^6, t = 1$, $\rho = 1.1, \mu_{max} = e^6$.
Repeat
1. Update $\boldsymbol{Z}$ by
   $\boldsymbol{Z}_{t+1} = \arg\min \|\boldsymbol{Z}_t\|_{w,*} + \frac{\mu_t}{2}\|\boldsymbol{Z}_t - (\boldsymbol{X} - \boldsymbol{E}_t - \boldsymbol{G}_t + \frac{\boldsymbol{Y}_1^t}{\mu_t})\|_F^2$.
2. Update $\boldsymbol{E}$ by
   $\boldsymbol{E}_{t+1} = \arg\min \alpha\|\boldsymbol{E}_t\|_{\tilde{w},*} + \frac{\mu_t}{2}\|\boldsymbol{E}_t - \frac{1}{2}(\boldsymbol{A}_t + \boldsymbol{B}_t)\|_F^2$,
   where $\boldsymbol{A}_t = \boldsymbol{X} - \boldsymbol{Z}_{t+1} - \boldsymbol{G}_t + \frac{\boldsymbol{Y}_1^t}{\mu_t}$, $\boldsymbol{B} = \boldsymbol{P}_t - \frac{\boldsymbol{Y}_2^t}{\mu_t}$
3. Update $\boldsymbol{G}$ by
   $\boldsymbol{G}_{t+1} = \arg\min \beta\|\boldsymbol{G}_t\|_1 + \frac{\mu_t}{2}\|\boldsymbol{G}_t - (\boldsymbol{X} - \boldsymbol{Z}_{t+1} - \boldsymbol{E}_{t+1} + \frac{\boldsymbol{Y}_1^t}{\mu_t})\|_F^2$.
4. Update $\boldsymbol{P}$ by
   $\boldsymbol{P}_{t+1} = \arg\min \gamma\|\boldsymbol{W} \cdot \boldsymbol{P}_t\|_1 + \frac{\mu_t}{2}\|\boldsymbol{P}_t - (\boldsymbol{E}_{t+1} + \frac{\boldsymbol{Y}_2^t}{\mu_t})\|_F^2$.
5. Update $\boldsymbol{Y}_1$ by
   $\boldsymbol{Y}_1^{t+1} = \boldsymbol{Y}_1^t + \mu_t(\boldsymbol{X} - \boldsymbol{Z}_{t+1} - \boldsymbol{E}_{t+1} - \boldsymbol{G}_{t+1})$.
6. Update $\boldsymbol{Y}_2$ by
   $\boldsymbol{Y}_2^{t+1} = \boldsymbol{Y}_2^t + \mu_t(\boldsymbol{E}_{t+1} - \boldsymbol{P}_{t+1})$.
7. Update μ by
   $\mu_{t+1} = \rho * \mu_t$.
8. Check convergence conditions:
   $\|\boldsymbol{X} - \boldsymbol{Z}_{t+1} - \boldsymbol{E}_{t+1} - \boldsymbol{G}_{t+1}\|_\infty < \varepsilon$ *and* $t < maxStep$.
9. Update $t = t + 1$.

Until convergent
Output: Defect matrix $\boldsymbol{E}$, defect-free matrix $\boldsymbol{Z}$.

3) Updating $\boldsymbol{G}$: Similarly, the optimization problem of $\boldsymbol{G}$ is

$$\boldsymbol{G}^* = \arg\min_G \beta\|\boldsymbol{G}\|_1 + \frac{\mu}{2}\left\|\boldsymbol{G} - (\boldsymbol{X} - \boldsymbol{Z} - \boldsymbol{E} + \frac{\boldsymbol{Y}_1}{\mu})\right\|_F^2. \tag{32}$$

According to [23], the optimization problem of (32) has a closed-form solution, which can be described as

$$\boldsymbol{G} = S_{\frac{\beta}{\mu}}\left(\boldsymbol{X} - \boldsymbol{Z} - \boldsymbol{E} + \frac{\boldsymbol{Y}_1}{\mu}\right) \tag{33}$$

where $S_{(\beta/\mu)}(x) = \text{sgn}(x)\max(|\ x\ | - (\beta/\mu), 0)$ is the soft thresholding operator.

4) Updating $\boldsymbol{P}$: Similarly, for variable $\boldsymbol{P}$, we have

$$\boldsymbol{P}^* = \arg\min_P \gamma\|\boldsymbol{W} \cdot \boldsymbol{P}\|_1 + \frac{\mu}{2}\left\|\boldsymbol{P} - \left(\boldsymbol{E} + \frac{Y_2}{\mu}\right)\right\|_F^2. \tag{34}$$

The optimization problem in (34) can be solved by the same way as in [21]. Finally, the proposed WDLRD can be solved by the iterative algorithm shown in Algorithm 1.

E. Convergence Analysis

In this section, we present a convergence analysis for the proposed Algorithm 1. According to the steps in Algorithm 1, there are mainly four suboptimization problems (i.e., Steps 1–4). For Steps 3 and 4, since they are norm-based optimization problems which are convex, they can be optimized easily. In this case, only the convergence of Steps 1 and 2 need to be proved. Due to Steps 1 and 2 are WNNM-based problems, we have a corollary as follows.

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Corollary 1: If the singular values satisfy $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n \geq 0$ and the weights satisfy $0 \leq w_1 \leq w_2 \leq \cdots \leq w_n$, then the global optimum of the WNNM problem is $d_i = \max(d_i - (w_i/2), 0)$.

Proof: The proof is the same as in [32].

For the proposed method, it penalizes larger singular values shrunk less to preserve the most important information, while smaller singular values shrunk more to remove unnecessary information. This satisfies the condition in Corollary 1. Thus, Corollary 1 guarantees that we have closed-form optimal solution for the proposed algorithm, which is a WNNM-based problem. Furthermore, we have the following theorem for the proposed Algorithm 1.

Theorem 1: If the weights are sorted in nondescending order, then the sequences $\boldsymbol{Z}_t$, $\boldsymbol{E}_t$, and $\boldsymbol{G}_t$ generated by Algorithm 1 satisfy

$$(1)\ \lim_{t\to\infty} \|\boldsymbol{Z}_{t+1} - \boldsymbol{Z}_t\|_F = 0.$$
$$(2)\ \lim_{t\to\infty} \|\boldsymbol{E}_{t+1} - \boldsymbol{E}_t\|_F = 0.$$
$$(3)\ \lim_{t\to\infty} \|\boldsymbol{G}_{t+1} - \boldsymbol{G}_t\|_F = 0.$$
$$(4)\ \lim_{t\to\infty} \|\boldsymbol{X} - \boldsymbol{Z}_{t+1} - \boldsymbol{E}_{t+1} - \boldsymbol{G}_{t+1}\|_F = 0.$$

Proof: Since the proof is similar to [32, Th. 2], we omit it because of space limitation.

Theorem 1 guarantees the convergence of Steps 1 and 2 in Algorithm 1. Therefore, the convergence of Algorithm 1 is guaranteed as we have proved the convergence for all the suboptimization problems in Algorithm 1.

F. Complexity Analysis

Suppose the size of a fabric image is $p \times q$, let $d = p \times q$. The computational complexity of Steps 1 and 2 in Algorithm 1 are $O(\max(pq^2, p^2q))$ respectively, which are related to SVD of a matrix with size of $p \times q$. If $p = q$, then the computational complexity of Steps 1 and 2 are $O(d^{1.5})$, respectively [35]. For Steps 3–6, the computational complexity are $O(pq)$, i.e., $O(d)$, respectively. Therefore, the total computational complexity of the Algorithm 1 is $O(2K(d^{1.5} + 2d))$ where K is the number of iterations times.

G. Comparison and Discussion

RPCA is applied in many applications as it can recover a low-rank matrix as the background and a sparse matrix as the noise or outliers from images. RPCA supposes that the recovered low-rank matrix is not sparse, and the acquired sparse matrix is not low rank. However, for specific fabric images, the defects are usually either sparse or low rank or even both of them. Therefore, RPCA may not be suitable in this case. For WNNM-RPCA, it was first proposed for image denoising by recovering the latent clean images from noisy images. WNNM-RPCA is more flexible than the NNM-based RPCA as it can apply various soft-thresholding values to choose important matrix singular values in SVD. In particular, WNNM can obtain better performance when proper weights or weighting rules are designed based on the prior knowledge or understandings of real problems [24]. However, the defect detection performance of WNNM-RPCA cannot be guaranteed since it is proposed mainly for image denoising in which the textures or the regularity are usually corrupted by discrete Gaussian noise, but not continuous defects with various shapes.

There are many other methods based on low-rank decomposition, such as Garbor + HOG (GHOG) method in [9] and weighted low-rank reconstruction (W-LRR) method in [10]. In [9], a direction-aware descriptor is designed for defect detection on patterned fabric based on the assumption that defect-free regions have specified direction, while defects will destroy the regularity of the directions. Low-rank decomposition is adapted to GHOG to recover the background and locate defects from the corrupted fabric images. The method is different from our method since it contains two-steps: feature extraction and defect locating. For our method, it is not necessary to extract features first from the fabric images. In addition, the defect prior to the fabric images is considered to guild defect detection instead of considering the directions of the fabric as in GHOG. Compared with W-LRR in [10], even though the defect prior is used in both our method and W-LRR, the construction of the prior map is different. The construction of the prior map in W-LRR is more complicated while our prior map is simple and easily calculated. In addition, the low-rank decomposition models are quite different. The objective function of W-LRR is to reconstruct the background by detecting defects from the fabric images, while our method focuses on improving the detection accuracy and precision by regularizing the background and the defect matrices to be low-rank with different weights. Therefore, the detected defect shapes of W-LRR and our method would be different.

IV. EXPERIMENTS

In this section, three databases are used to evaluate the performance of the proposed method. The first one is fabric images with regular patterns, the second one is unpatterned fabric images, and the third one is woven fabric images. Several related methods are selected to compare with the proposed WDLRD. These methods are RPCA [20], PNRPCA [21], saliency detection model based on wavelet transform (WT) [36], fabric defect inspection using prior knowledge guided least squares regression (PGLSR) [22], fabric inspection based on the Elo rating method (Elo) [37], and deformable patterned fabric defect detection with the Fisher Criterion-based deep learning method (SDA) [2]. In addition, our method using NNM [named Ours_NNM, i.e., the first two terms in (13) are based on the traditional nuclear norm instead of weighted nuclear norm] is also implemented and used as a compared method. The parameters of the compared methods are set like that in the original articles. The experiments are implemented in an Intel Core i7-8700K 3.70-GHZ CPU (with six cores and twelve threads), Geforce RTX 2080 environment, using MATLAB 2018a as the programming software.

Evaluation Metrics: In our experiments, a group of metrics are used to evaluate the performance of the proposed WDLRD and the compared methods. The metrics include recall, false alarm, precision (PR), accuracy (ACC), and F-measure. The

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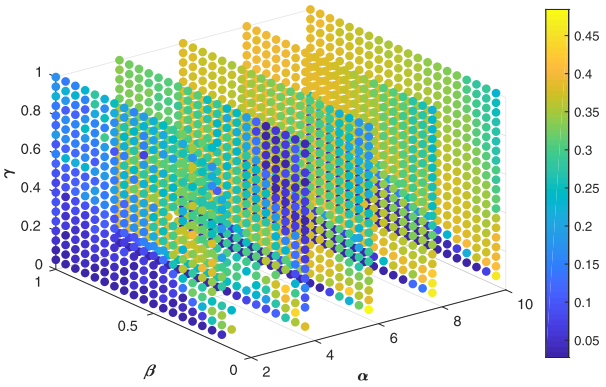


Fig. 5. F-measure versus various parameters α, β, and γ on HKBU database.

calculating formula of these metrics are based on statistical parameters, including true positive (TP), false positive (FP), true negative (TN), and false negative (FN) [38]

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{35}$$

$$\text{false alarm} = \frac{\text{FP}}{\text{FP} + \text{TN}} \tag{36}$$

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{37}$$

$$F\text{-measure} = 2\frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}. \tag{38}$$

Parameters Analysis: To obtain better performance of the proposed method, three factors are needed to consider. The first one is the patch size of fabric images while constructing the prior map. Similar to [21] and [22], the patch size is 16×16 with 8×8 overlapping.

Second, there are three parameters in the proposed objective function, and we need to find the optimal values for these parameters. In our experiments, the grid search strategy was used to find the optimal values of the parameters and the variation areas of α, β, and γ were $[2, 4, \ldots, 10]$, $[0.001, 1]$ and $[0.001, 1]$, respectively. The average F-measure versus different values of α, β, and γ on the dot-, box-, and star-patterned fabric of the Hong Kong Baptist University (HKBU) database are shown in Fig. 5. In Fig. 5, the values of F-measure are presented with different colors corresponding to the color bar on the right-hand side. The result shows that when the values of α, β, and γ are set properly, the proposed method is robust and obtains better performance. Especially, it would be better to set the values of α much larger than that of β and γ in order to penalize the defect matrix to be low rank. In addition, it is not necessary to set a large penalty on the noise term when there is no heavy noise interference on the fabric image, and a small penalty should be given to the prior term to avoid misleading when the prior tends to be incorrect. With such knowledge, we set $\alpha = 6.00$, $\beta = 0.053$, and $\gamma = 0.053$ for all databases for simplicity. The positive constant C in (25) and (31) is set as 1 in all experiments.

Lastly, we need to decide a threshold to convert the saliency map to a binary map. The method used in our experiments is Otsu's method [39], which chooses the element that has the maximal variance between classes as the threshold automatically. The Otsu method is also used by other compared methods. For Ours_NNM, the optimal values of α, β, and γ are set as 4, 0.05, and 0.06, respectively. In all experiments, the defect detection results are presented in Figs. 6–14, while white pixels represent defective regions, and the black pixels stand for the background.

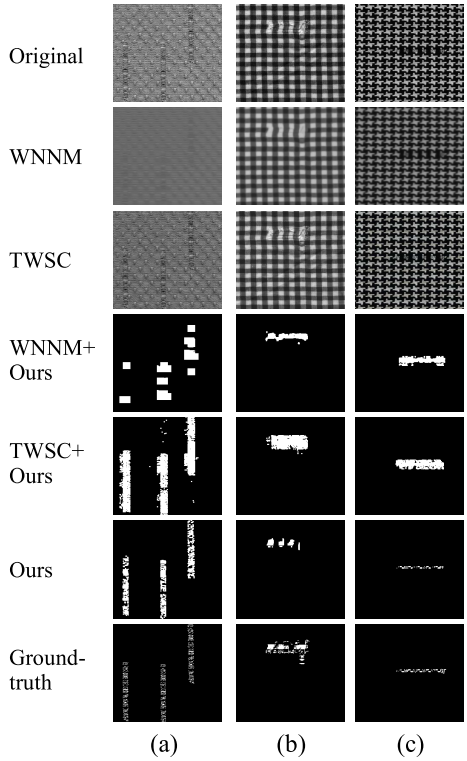


Fig. 6. Examples showing the detection results on (a) dot-, (b) box-, and (c) star-patterned fabric images.

A. Experiments on the HKBU Fabric Image Database

The HKBU fabric image database is from the Research Associate of Industrial Automation Research Laboratory, HKBU [40]–[42]. It contains three types of fabric images, i.e., dot-, box-, and star-patterned fabric images. There are a total of 60 dot-patterned and 50 star-patterned fabric images. Half of these two kinds of images are defect free, and the others are defective. The box-patterned fabric data set contains a total of 56 images, of which 30 images are defect free. All images are in the same size of 256×256 pixels. There are ground-truths for each defective image, while defective regions are presented as white, and the defect-free regions are black. The sample images on this database are shown in Fig. 15.

1) Comparison of Different Methods and the Proposed Method: In this experiment, the defect detection results on dot-patterned, box-patterned, and star-patterned fabric images are shown in Figs. 8–10, respectively. In Figs. 8–10, the first row is the original image, the second to the last row are RPCA, PNR-PCA, WT, PGLSR, Elo, SDA, the proposed WDLRD based on NNM (Ours_NNM) and WNNM (Ours), the corresponding prior map used by our method, and the ground-truth images, respectively. From Fig. 8, we can see that RPCA cannot

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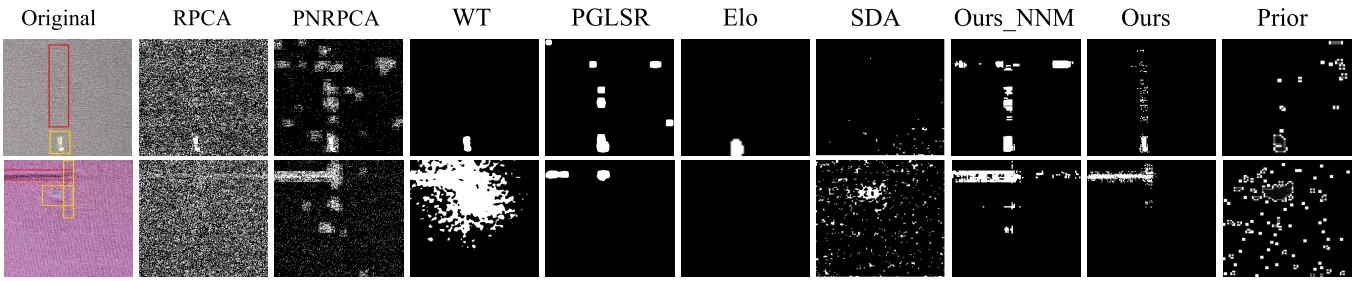


Fig. 7. Detection results of different methods in the case that multiple types of defects are present in a fabric image. The first row: netting multiple (red bounding box) and broken end (yellow bounding box); the second row: folding (red bounding box) and broken end (yellow bounding box).

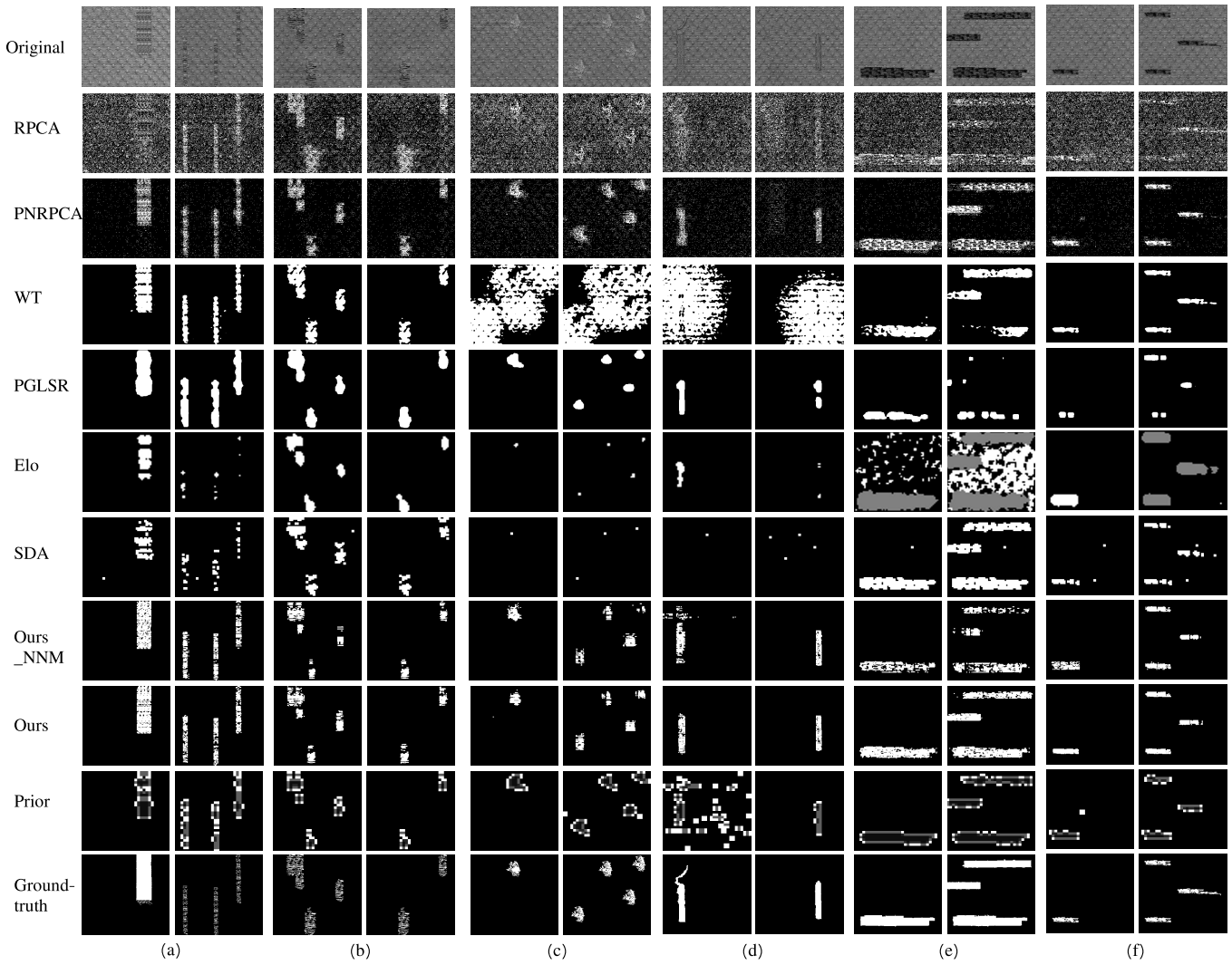


Fig. 8. Detection results for dot-patterned fabric images. The defect types are (a) broken end, (b) hole, (c) knots, (d) netting multiple, (e) thick bar, and (f) thin bar.

precisely distinguish the defective regions and the background as their binary maps are not distinct. The possible reason is that RPCA tends to generate fake defects when the fabric patterned images have noise or illumination variations. WT is able to locate the defective regions in many cases, especially on dot-patterned fabric images. As shown in Fig. 8(f), when the defect type is a thin bar, the performance of WT is the best. However, in other cases, such as knots in Fig. 8(c), or on box- or star-patterned fabric images, the detection results of WT are strange. As we can see, its detected defective regions are much larger than the real-defective regions. The reason for this phenomenon is that WT is a method based on wavelet transform, which makes it very sensitive to the variation of illumination and gray level of the images.

Compared with WT, PGLSR is not very sensitive even though it cannot locate defective regions as precise as WT

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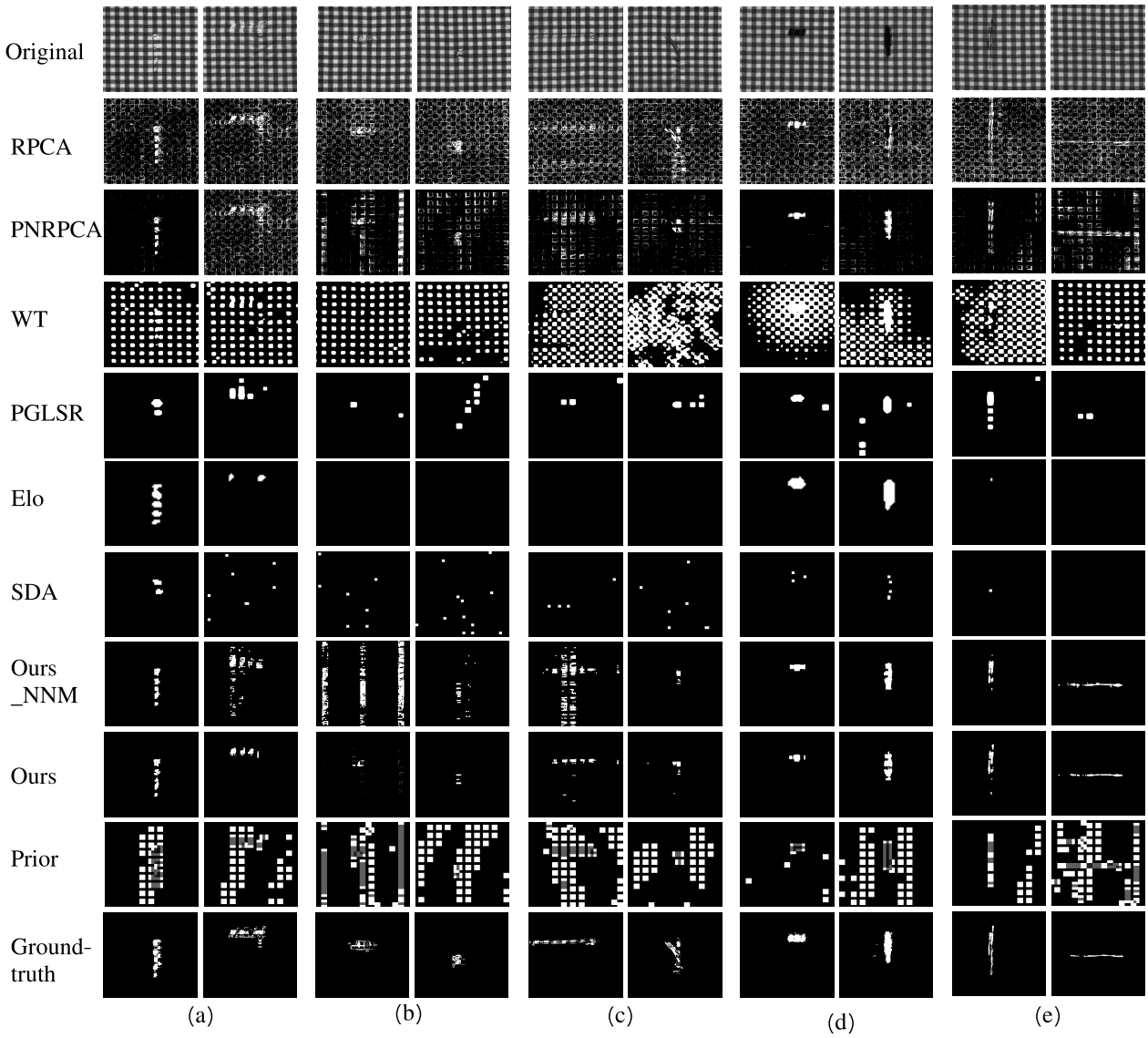


Fig. 9. Detection results for box-patterned fabric images. The defect types are (a) broken end, (b) hole, (c) netting multiple, (d) thick bar, and (e) thin bar.

in several cases [Fig. 8(b) and (f)]. Based on the detection results of PGLSR, we can see that PGLSR tends to generate inaccurate shapes for real defective regions due to the similarities between the texture of defects and background.

Different from the compared methods, the proposed WDLRD is more robust, as we can see from Figs. 8–10, that the defected defective regions by WDLRD are always smaller or even similar to the ground truth. Even though WDLRD and PNRPCA and PGLSR use the similar prior map to guide the detection, their final detections results are not the same. The weighted low-rank technique contributes to the good performance of WDLRD, and thus, is the most important factor to guarantee WDLRD to obtain more precise detection results than PNRPCA and PGLSR.

Besides visual defect locating results, the metrics of false alarm, recall, precise, accuracy, and F-measure are also used to quantify the locating accuracies of the proposed WDLRD and other compared methods. Tables I–III list the locating accuracy while Figs. 16 and 17 show the false alarm rate-recall graphs and the F-measures for the three-patterned fabric images. The average false alarm, recall, precision, accuracy, and F-measures listed in Tables I–III are obtained by calculating the average of the false alarm, recall, precision, accuracy, and F-measure of each image. In Fig. 16, each point represents the false alarm versus recall of an image.

From Table I, we can see that the false alarm of the proposed method is small, and the precision, accuracy, and F-measure are the highest compared with other methods. This indicates that the proposed WDLRD does not tend to generate false positive detection results, and the detected regions in terms of no matter defects or background are much more precise than that of the other methods. For WT, even though its recall is the highest, its false alarm rate and precision and accuracy are not very satisfactory because it is sensitive to the lighting conditions and gray level changes. In Fig. 16, points lie on the left top side indicate more accurate detection results as we expect

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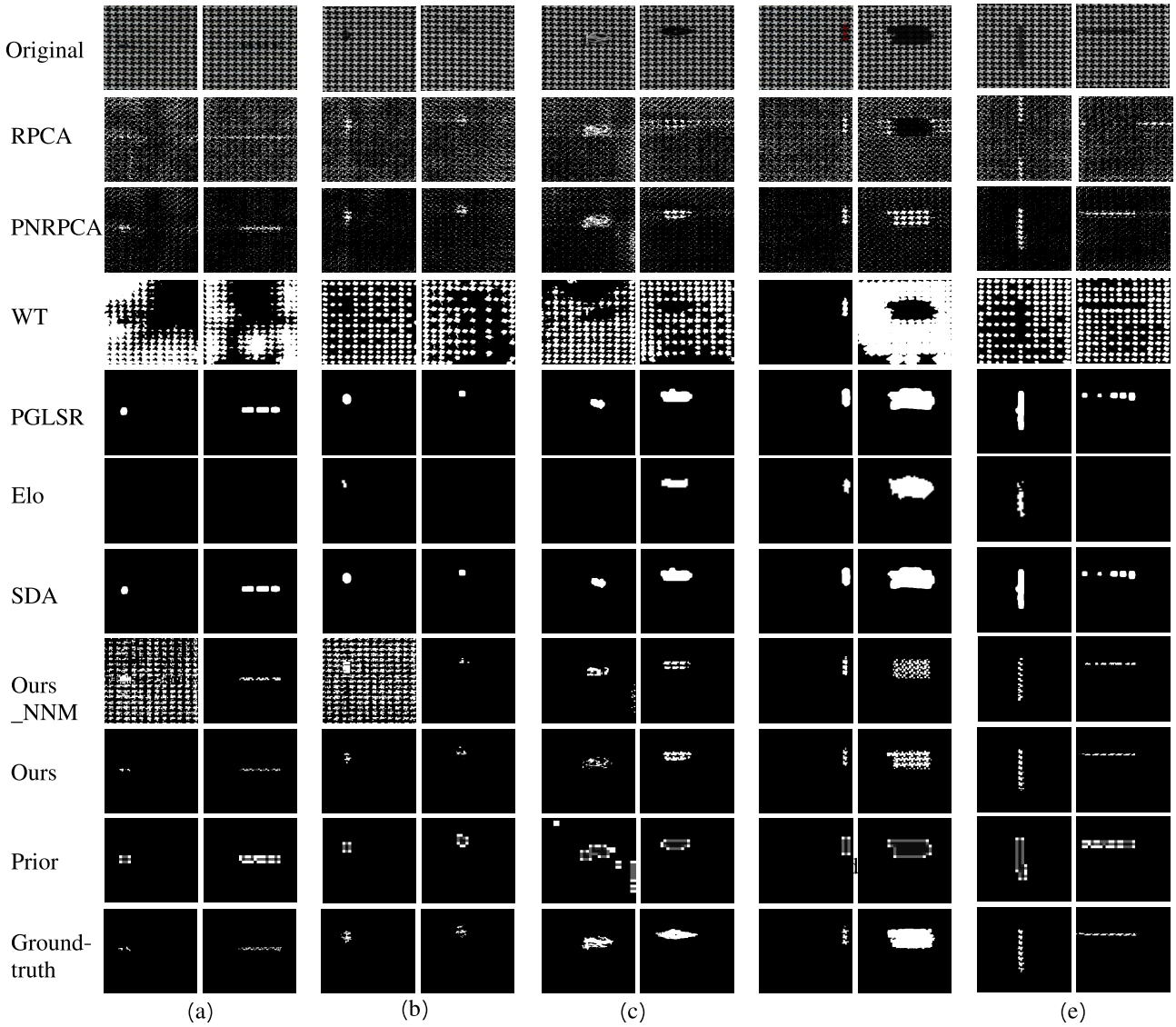


Fig. 10. Detection results for star-patterned fabric images. The defect types are (a) broken end, (b) hole, (c) netting multiple, (d) thick bar, and (e) thin bar.

the false alarm to be as small as possible while the recall to be as large as possible. From the three graphs, we can see that the points provided by our method lie on the left top side in most cases. Even though several red points lie on unsatisfactory positions, as shown in Fig. 16(b), we can still conclude that the overall performance of locating defective regions of the proposed method is better than the compared methods. This conclusion can also be verified by the F-measure, as shown in Fig. 17.

2) Denoising Separately Versus Noise Handling Incorporating in the Proposed Method: In this section, we explore the performance of all methods in defect detection after denoising. Specifically, WNNM [24] and trilateral weighted sparse coding (TWSC) [43] are used to preprocess the noise of the defective images on HKBU database, and then the compared methods and the proposed method without noise term [i.e., $\boldsymbol{G}$ is removed from the objective function in (13)] are used to locate defect regions. For simplicity, we name our method under the two denoising environments as WNNM + Ours and TWSC + Ours, respectively.

The detection results of using WNNM and TWSC on the HKBU database are shown in Figs. 18 and 19, respectively. By comparing the results of all methods in Figs. 17–19, we can know that the performance of both RPCA and pn-RPCA after denoising is better than that without denoising, while WT obtains lower performance. The result of TWSC + Ours is better than that of WNNM + Ours, but they are both lower than Ours. The reason is that using WNNM separately before our method will destroy the intrinsic difference between the defect and the background. As we can see from the second row of Fig. 6, especially the second image in Fig. 6(a), the distinction between the defect region and the background is very blurry after denoising. This will affect the precision of later defect detection using the proposed method as the main advantage of our method is that it can treat the matrix singular values differently via assigning different weights, by which the most important/distinguishing characteristics of

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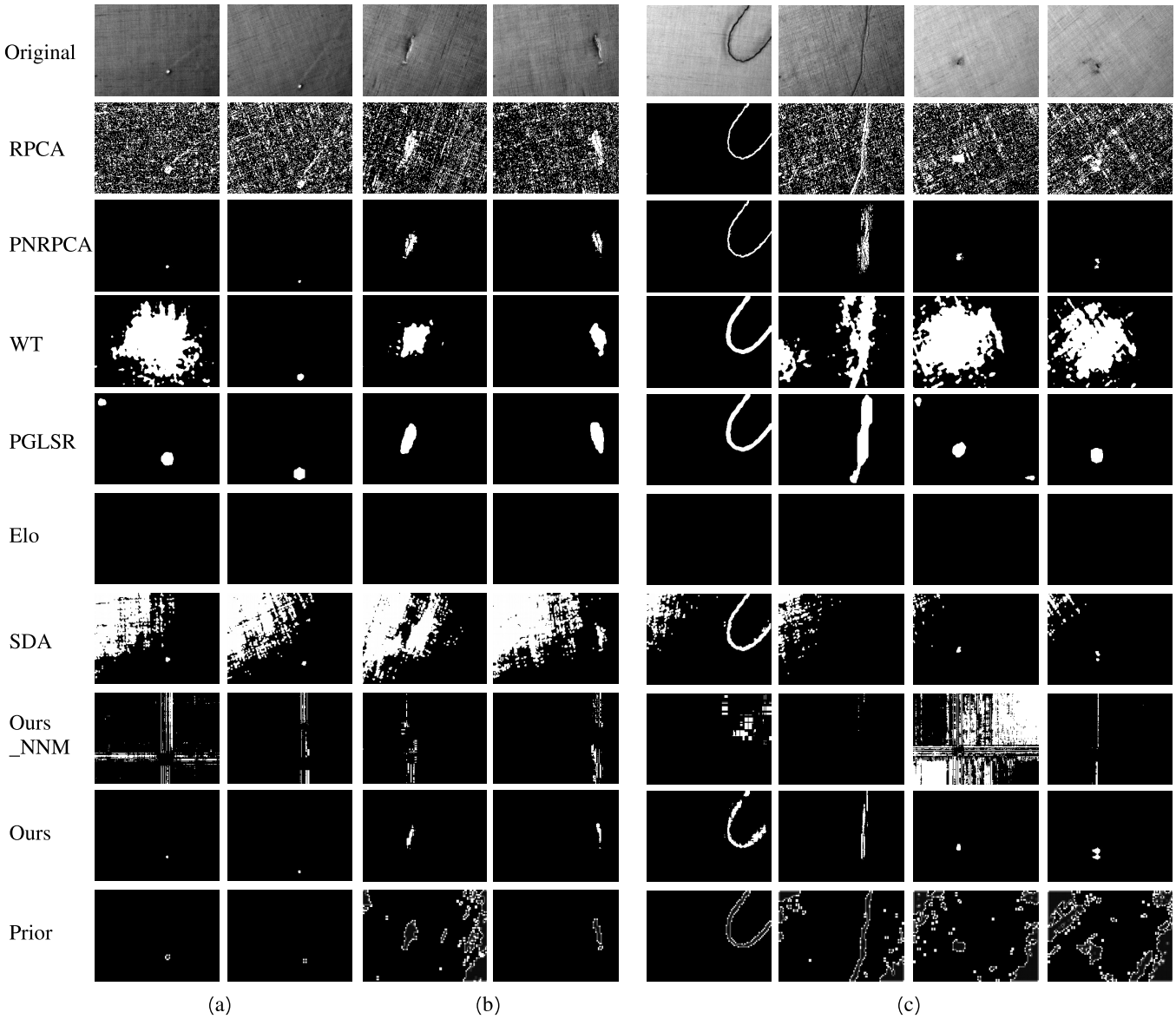


Fig. 11. Results of defect locating on plain fabrics on TILDA database. The defect type is (a) hole, (b) broken end, and (c) sports.

a fabric image can be preserved. It can also be verified by the results of TWSC + Ours in the third row in Fig. 6 as TWSC preserves most of the important/distinguishing characteristics of the fabric images, and thus, the detection result of TWSC + Ours in Fig. 19 is better than that of WNNM + Ours in Fig. 18.

Therefore, we can conclude that if separate denoising removes the distinguishing characteristics of the fabric images, and the later detection performance of the proposed method would be affected. If we use suitable denoising methods, the proposed method can still be more effective than other compared methods. However, incorporating noise handling function into the proposed optimization model is a better way as it can obtain better defect detection performance and is easy to optimize in an end-to-end manner.

B. Experiments on the TILDA Database

TILDA is a Textile Texture database [44]. It contains eight representative textile types, including seven defective classes and one defect-free class. There are a total of 3200 images on this database. In our experiment, two different types of plain fabric (total of 250 fabric images) were used to evaluate the effectiveness of the proposed WDLRD. The size of the original images is 512×768 pixels, and they were downsized to 171×256 in our experiment. The defective samples of the two types of fabric are shown in Fig. 20.

Due to the lack of ground-truth images on the TILDA database, the quantitative evaluations, such as precision and accuracy, are not calculated. Visualization results on the defective region detection are presented in Figs. 11–13. From the results, we can see that the proposed method is effective in case that the defects are in the shape of the hole, dot, and block. However, it is not suitable for the situation when defects are long and zigzag in diagonal or even horizontal/vertical direction. The detective regions obtained by WT are very interesting as they explode like fireworks. The binary maps obtained by PNRPCA still contain many noises while

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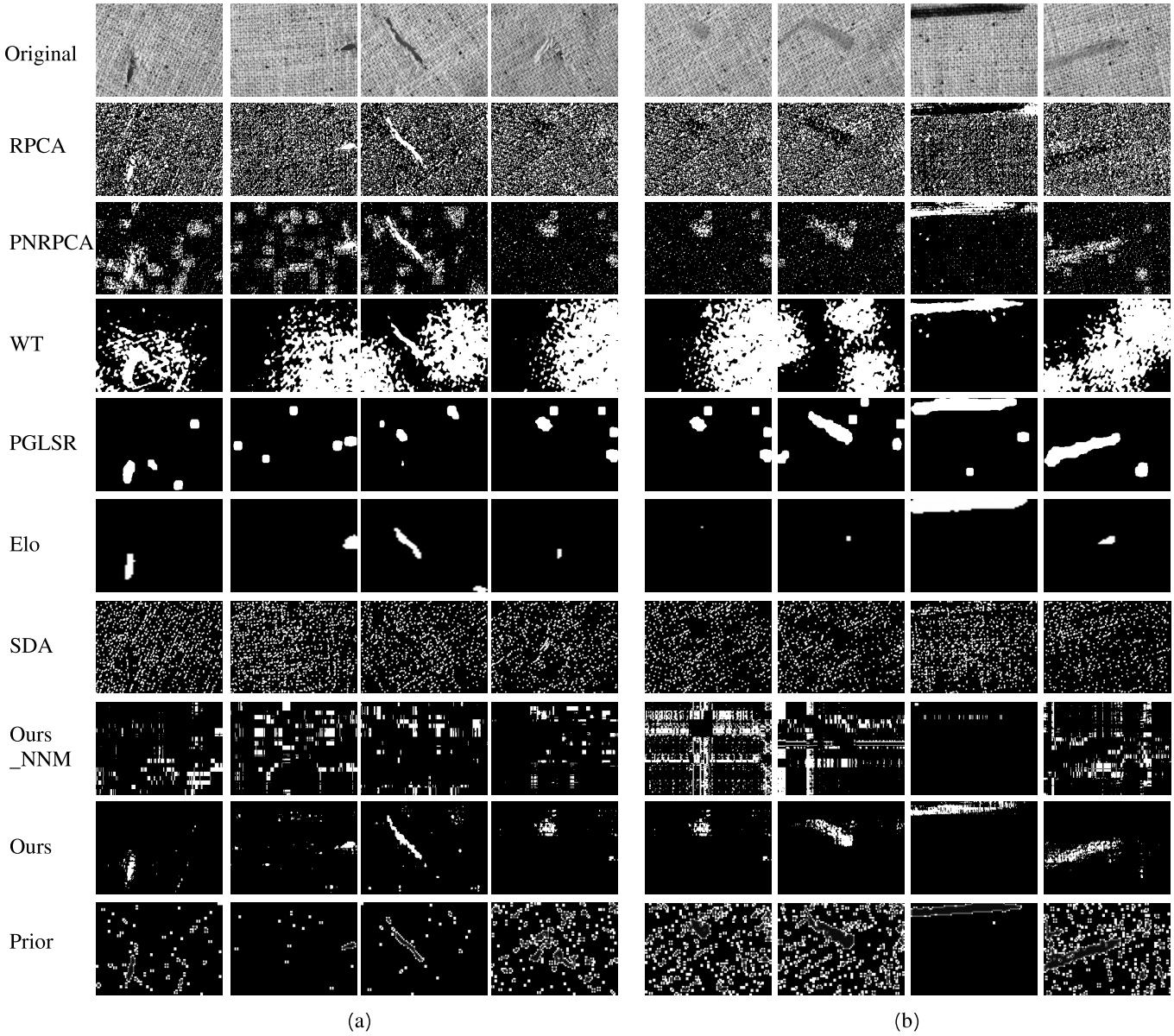


Fig. 12. Results of defect locating on plain weave fabrics on TILDA database. The defect types are (a) broken end and (b) stain.

the binary maps obtained by our methods are very clear, which indicates that the noise term based on F-norm in PNRPCA is not as effective as that based on L_1-norm in our method.

C. Experiments on the TAAI Database

In this section, we use the fabric images that collected by our research group named Textile & Apparel Artificial Intelligence (TAAI) Research Group of Institute of Textiles and Clothing (ITC) in The Hong Kong Polytechnic University, Hong Kong. The images were captured and resized to 256×256 pixels from a textile inspection machine in a textile factory by an industrial camera (BB-500GE). The database contains 500 woven fabric images, of which half are defective images, while the others are defect-free images. The defects were divided into five typical categories, namely holes, jump yarn, stain, float, and netting multiple. The samples are shown in Fig. 21. From our observation, the defects on this database are often of hole or block shapes, which is a suitable application scenario for the proposed method. In particular, for a defective image on the TAAI database, when the defects are in hole or block forms, it is instinctive that the backgrounds are composed of homogeneous textures, and the defects are also composed of continuous and homogeneous components. Thus, both the background matrix and the defect matrix in such cases are low rank. This is consistent with the property of the proposed method. Therefore, it is necessary to evaluate the performance of the proposed method of fabric defect detection in this database.

As can be seen from Fig. 21, the woven fabric contains regular hollowing out, which makes the detection of defects

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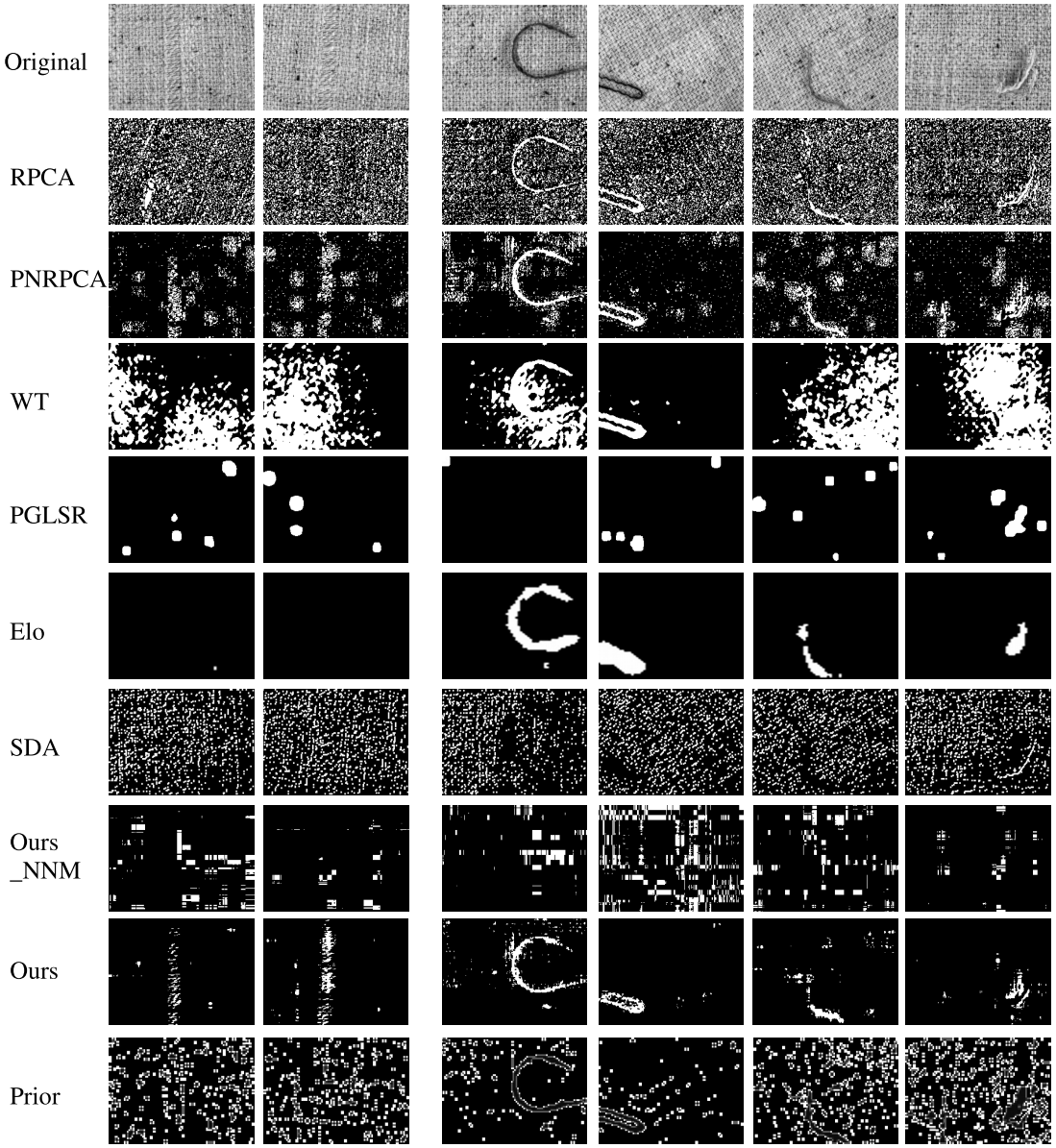


Fig. 13. Results of defect locating on plain weave fabrics on TILDA database. The defect types are (a) netting multiple and (b) jump yarn.

more challenging, especially when the defects are not distinct. The locating results of defects are shown in Fig. 14. From the results, we can see that the locating performance of all methods is not satisfactory. PGLSR tends to locate inaccurate regions when the gray of the stain is similar to that of the fabric. The detection results of PNPCA contain many noises, while RPCA cannot highlight the defects in a distinct manner. WT is sensitive to the pattern of the background. The proposed method can locate the defective regions roughly even though the shapes of the detected defective regions are not very similar to that of the true defects.

From the experiments on the HKBU database, the TILDA database, and the TAAI database, we found that the proposed algorithm is able to converge to a local minimum after about 100 iterations when conducting defect detection for every image. The convergence curves corresponding to images on the three databases are similar, and we report only one among them in Fig. 22 as examples. From Fig. 22, we can intuitively see the variations of the objective function value versus the increasing iteration times. The convergence curves indicate that the proposed objective function finally achieves optimal local solution by using the ADMM optimization algorithm.

The computational time of different methods is shown in Table IV. The result shows that the computational time of most methods on a personal computer is less than 1 s and the proposed method can locate defects in less than 0.3 s. Furthermore, the proposed method will definitely be more efficient on a high-performance computer. Therefore, it is reasonable to conclude that the computational time of the proposed method is acceptable for the application of real-time fabric inspection.

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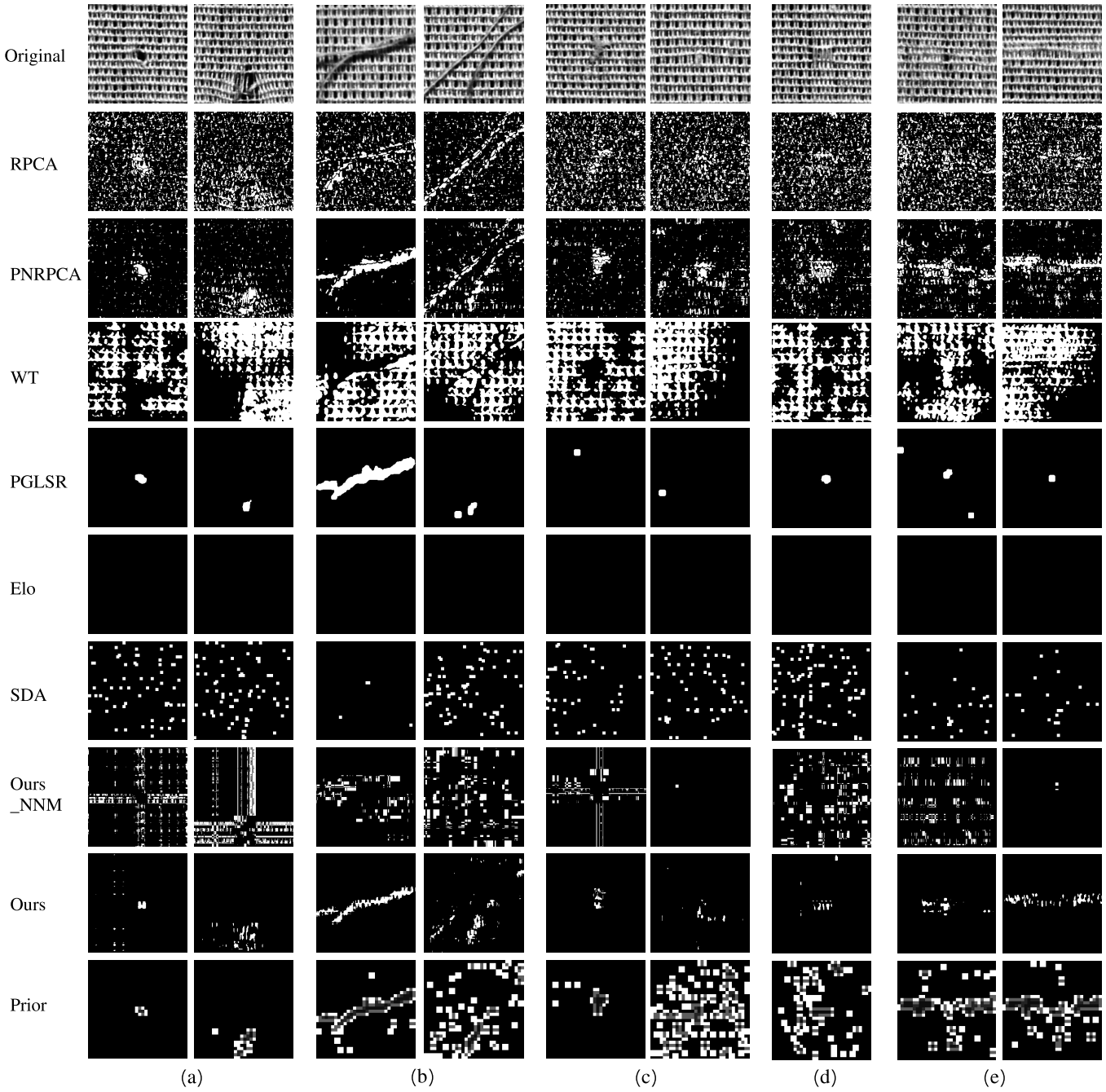


Fig. 14. Results of defect locating on TAAI fabric database. The defect types are (a) holes, (b) jump yarn, (c) stain, (d) float, and (e) netting multiple.

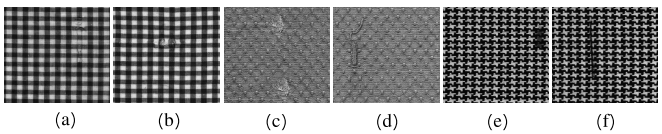


Fig. 15. Defective images on the HKBU database. (a) Broken end. (b) Hole. (c) Knots. (d) Netting multiple. (e) Thick bar. (f) Thin bar.

TABLE I
AVERAGE LOCATING RESULTS FOR DOT-PATTERNED FABRIC IMAGES

Method	False Alarm	Recall	PR	ACC	F-measure
RPCA	0.27	0.59	0.10	0.71	0.17
PN-RPCA	0.04	0.65	0.42	0.93	0.51
WT	0.20	0.82	0.43	0.80	0.56
PGLSR	0.02	0.58	0.66	0.95	0.62
Elo	0.07	0.49	0.48	0.91	0.49
SDA	0.02	0.47	0.54	0.96	0.50
Ours_NNM	0.01	0.66	0.65	0.96	0.66
Ours	0.02	0.73	0.67	0.97	0.69

D. Cases Study on Multiple Types of Defects Present on a Fabric Image

In this experiment, we explore the performance of the proposed method in the case that multiple types of defects are present in a fabric image. The fabric images in this experiment are selected from the Tianchi database that is obtained from the

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TABLE II
AVERAGE LOCATING RESULTS FOR BOX-PATTERNED FABRIC IMAGES

Method	False Alarm	Recall	PR	ACC	F-measure
RPCA	0.19	0.61	0.04	0.81	0.08
PN-RPCA	0.05	0.47	0.22	0.94	0.3
WT	0.27	0.38	0.02	0.72	0.05
PGLSR	0.02	0.21	0.16	0.97	0.18
Elo	0.00	0.00	0.04	0.99	0.01
SDA	0.01	0.05	0.17	0.98	0.07
Ours_NNM	0.04	0.41	0.34	0.95	0.37
Ours	0.04	0.42	0.44	0.95	0.43

TABLE III
AVERAGE LOCATING RESULTS FOR STAR-PATTERNED FABRIC IMAGES

Method	False Alarm	Recall	PR	ACC	F-measure
RPCA	0.16	0.57	0.03	0.83	0.07
PN-RPCA	0.05	0.60	0.15	0.94	0.24
WT	0.37	0.14	0.03	0.63	0.04
PGLSR	0.01	0.51	0.39	0.99	0.45
Elo	0.00	0.17	0.20	0.99	0.18
SDA	0.00	0.05	0.14	0.98	0.08
Ours_NNM	0.07	0.45	0.49	0.92	0.47
Ours	0.02	0.58	0.54	0.98	0.56

AI Challenge in Xuelang Manufacturing: Visual Computing Assisted Quality Inspection in 2018 [45].

As shown in Fig. 7, the original images contain multiple types of fabric defects. For the first row, the defect region highlighted by the red bounding box in the original image is netting multiple, while that highlighted by the yellow bounding box is the broken end. For the second row, the defect region highlighted by the red bounding box in the original image is folding, while highlighted by the yellow bounding box is the broken end. As we can see from Fig. 7, even though the prior of the proposed method is not very accurate, the proposed method can finally locate the defective regions in a clear and accurate manner. The locating results of PNRPCA contain many noises, while the proposed method can locate the defective regions more clearly. The potential reason is that the proposed method utilizing L_1-norm as the penalty can reduce the noise interference while PNRPCA using L_2-norm on the noise term cannot provide sparsity, and thus, the final defection results still contain much noise as fake defects. For Ours_NNM and Ours, we can see that even though they both use the same prior map and noise term, the detection results are different. Our method based on WNNM can highlight the main defective regions and ignore the interference of fake defects by treating the matrix singular values differently via shrinking less oriented to large variation and shrinking more oriented to small variation of the fabric image, while Ours_NNM based on traditional NNM treats the matrix singular values equally, and thus, cannot highlight the main defective regions when the singular values shrink too much and locate fake defects when the singular values shrink too little (this observation can be verified by the experimental results on all databases).

E. Experimental Results and Discussion

Results of defect detection on the patterned and unpatterned fabric images and quantitative evaluations on patterned fabric images are presented in Section IV-D. Based on the overall experimental results, we have the following interesting points.

1) RPCA cannot distinguish the defective regions in a clear manner. In particular, the performance of defect detection relies on the lighting condition and gray level of the images. When there are various illumination and gray levels, RPCA tends to generate many noises and obtain low precision. Thus, RPCA provides ambiguous or even misleading and false decisions. The potential reason for the low performance is that RPCA does not take the prior knowledge of fabric texture into consideration and distinguish defects from a global perspective is not reliable when the contrast between background and the defect is not distinct.
2) Even though PNRPCA, PGLSR, and the proposed WDLRD all use the similar prior map to guide the defect detection, their performance is different in many cases. PGLSR tends to generate blocklike defective regions when the real defects are tiny and long, especially in the case when the defect type is broken or the appearance of defect is line shape. In many cases, the shapes of detected defective regions by PNRPCA approximate to that of the real-defective regions. However, there are many white pixels in the binary map, which leads to low accuracy for PNRPCA. In contrast, the proposed WDLRD can not only locate the real-defect region clearly but also distinguish the fake defects from the fabric images in most cases, even when the prior is not very accurate.
3) The performance of Ours_NNM is comparable to or even better than PNRPCA and PGLSR in most cases (it can be clearly verified by Fig. 17). This indicates the effectiveness of the proposed double-low-rank model. However, as we can see from all results, our method using weighted low-rank decomposition (Ours) performs better than that using traditional low-rank decomposition (Ours_NNM). It is reasonable to conclude that weighted low-rank decomposition can treat the matrix singular values differently by which the most important characteristics of defects or background of a fabric image can be preserved. With this property, the proposed method can locate defects in a clearer manner while Ours_NNM would ignore the importance of some defects. Especially, when the defective region is blurry, Ours_NNM cannot effectively locate the correct position of real defects and instead highlights the fake defects [this can be seen in Fig. 10(a) and (b)].
4) The detection results of Elo are empty in some cases. The potential reason is that Elo achieves the goal of defect detection by fair matches between partitions on the fabric images. It is implemented by dividing the images into partitions of standard size and then updating the matches between various partitions through the Elo point matrix. The detection performance of Elo is sensitive to the size of partitions, and that is why it obtains unsatisfactory performance in some cases.

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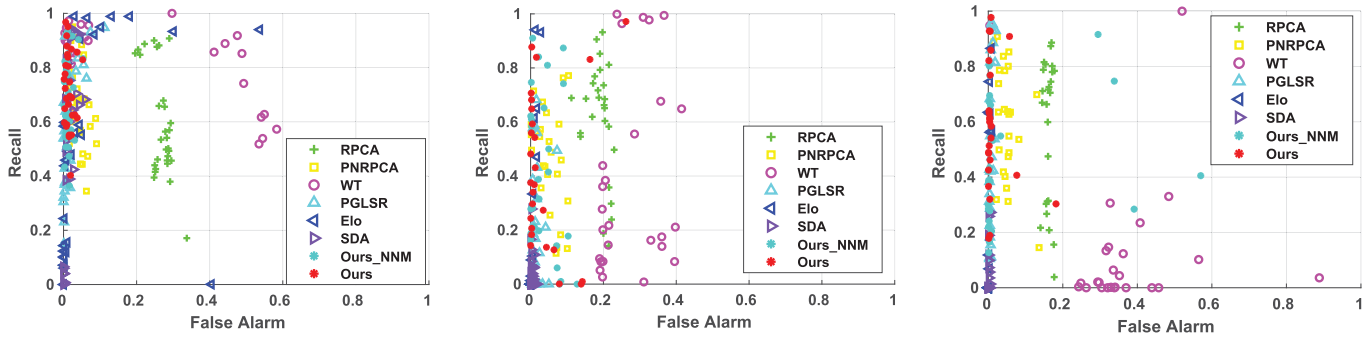


Fig. 16. False alarm-recall graph for dot-, box-, and star-patterned fabric images.

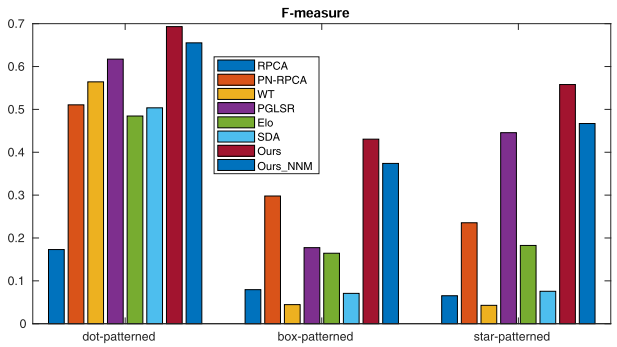


Fig. 17. F-measure of different methods on dot-, box-, and star-patterned fabric images of the HKBU database.

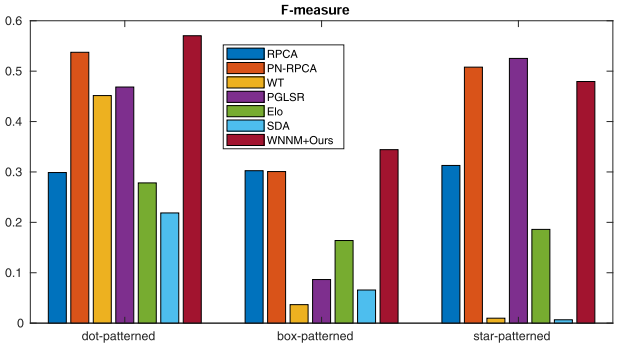


Fig. 18. F-measure of all methods after denoising with WNNM on dot-, box-, and star-patterned fabric images of the HKBU database.

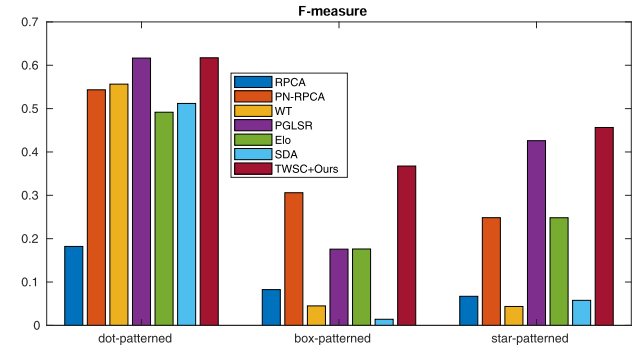


Fig. 19. F-measure of all methods after denoising with TWSC on dot-, box-, and star-patterned fabric images of the HKBU database.

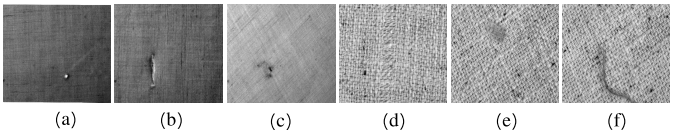


Fig. 20. Defective images on the TILDA database. (a) Hole. (b) Broken end. (c) Spots. (d) Netting multiple. (e) Stain. (f) Jump yarn.

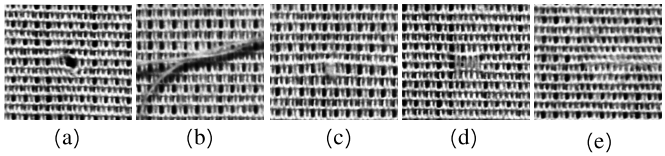


Fig. 21. Defective images on TAAI database. (a) Hole. (b) Jump yarn. (c) Stain. (d) Float. (e) Netting multiple.

The detection results of SDA seem to be scattered or continuously large areas. One reason for this phenomenon is that the procedure of locating defects by SDA involves dividing the fabric images into patches of the same size, which indicates that it is sensitive to the patch size. Since the patch size for SDA on all databases is set like that on the HKBU fabric image database in the original article, the performance of SDA on the other databases in our experiments cannot be guaranteed. Another reason is that SDA focuses on detecting defect for patterned fabric, and the performance on the nonpatterned fabric has not been explored yet.

5) In general, the proposed WDLRD can obtain higher accuracy and precision than other compared methods. However, its effectiveness is limited in the case that the defects are very tiny and long. Such characteristic is consistent with the property of the objective function of WDLRD. Since the defect matrix of WDLRD is supposed to be low rank, some real-defective regions are missed, and the final detected defective regions are shrunk to a block [such as the left image in Fig. 8(d) and the right image in Fig. 9(c)].

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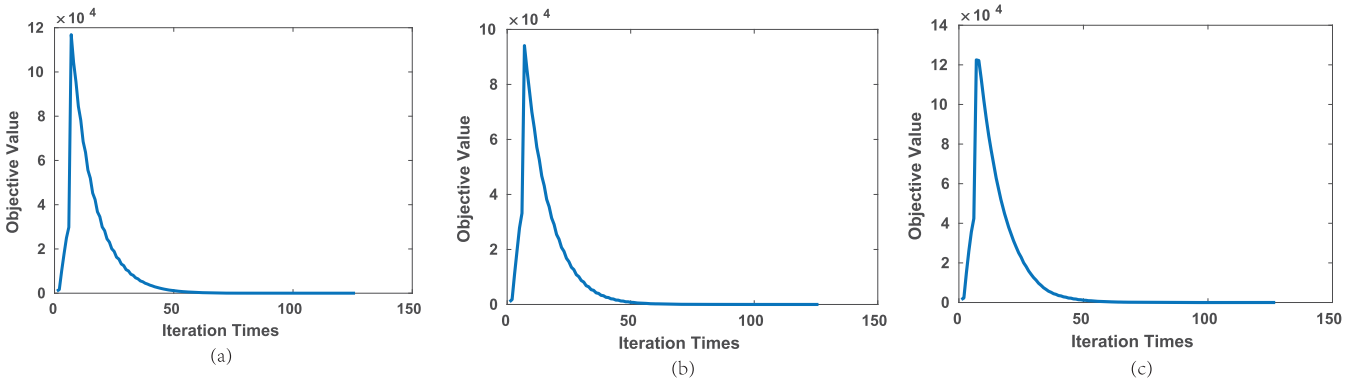


Fig. 22. Convergence curves on (a) HKBU database, (b) TILDA database, and (c) TAAI database.

TABLE IV
COMPUTATIONAL TIME OF DIFFERENT METHODS (UNIT: SECONDS/IMAGE)

Method	HKBU Database	TILDA Database	TAAI Database
Image Number (Size)	166 (256 × 256)	250 (256 × 171)	500 (256 × 256)
RPCA	0.0775	0.0360	0.0725
PNRPCA	0.0455	0.0250	0.0502
WT	0.1071	0.0707	0.1601
PGLSR	1.1432	0.4001	1.0110
Elo	Training Stage: 0.0414 Testing Stage: 0.0424	Training Stage: 0.0406 Testing Stage:0.0400	Training Stage: 0.0461 Testing Stage: 0.0395
SDA	Training Stage: 1.8943 Testing Stage:0.0767	Training Stage: 1.9228 Testing Stage: 0.0658	Training Stage: 1.8594 Testing Stage: 0.0783
Ours_NNM	0.1327	0.1118	0.1675
Ours	0.2812	0.1529	0.2786

V. CONCLUSION

In this article, we propose a WDLRD method for fabric defect detection. The proposed method regularizes the background matrix and defect matrix with a weighted nuclear norm, by which the matrix singular values can be shrunk differently to preserve the most important information. A defect prior map is constructed in the proposed model to guide the defect detection. In addition, a noise matrix is used to capture the noise or other interference on fabric images. We also propose an ADMM-based algorithm to solve the optimization problem. Experiments on three fabric databases were conducted to evaluate the defect detection performance of the proposed method. Experimental results indicate that the proposed method can obtain higher accuracy and lower false alarm rates on plain and regular patterned fabric images compared with the other methods.

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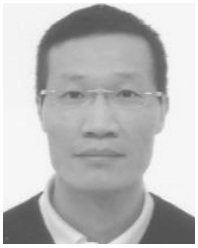
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